Measuring Smartphone Price Decline: How Modeling and Index Construction Shape Inflation Estimates

Dejia (Edgar) Zhu

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Abstract

High–technology markets such as smartphones feature rapid model turnover, frequent promotional discounting, and continual improvements in product quality. In such environments, traditional matched-model indices like the Jevons index often record steeper price declines than quality-adjusted alternatives because they treat temporary discounts, product entry and exit, and quality change differently.

This paper examines how alternative index methodologies measure smartphone price change between 2020 and 2025. Using longitudinal price data scraped from Amazon via the Keepa API and detailed specifications for major brands including Apple, Samsung, and Google, the study constructs traditional matched-model Jevons indices alongside several hedonic Jevons indices based on feature-adjusted price predictions. Hedonic models are estimated using regularized regressions to accommodate high-dimensional product characteristics.

Across both quarterly and annual frequencies, traditional indices consistently imply larger cumulative price declines than quality-adjusted alternatives. However, the magnitude of this divergence depends strongly on model design. Level-based hedonic specifications produce substantially smaller measured deflation, whereas price-change-based hedonic models generate price paths that closely track traditional indices despite adjusting for quality.

By comparing matched-model, level-based hedonic, and price-change hedonic approaches within a unified framework, this study highlights how econometric specification shapes measured inflation in rapidly evolving product markets and provides guidance for constructing price indices for high-technology goods.

1 Introduction

The smartphone market has undergone continual transformation over the past decade, driven by rapid technological innovation, short product cycles, and increasingly sophisticated consumer expectations. Each year, manufacturers such as Apple, Samsung, and Google introduce new models featuring faster processors, higher-resolution cameras, expanded memory, and improved displays. These advances stimulate frequent consumer upgrades while simultaneously accelerating the depreciation of older models, particularly as devices approach end-of-life promotions and clearance pricing (Bayus, 1991; Shocker et al., 2004).

These market dynamics present significant challenges for measuring constant-quality price change. Using evidence from high-technology categories, the Bureau of Labor Statistics (2024) documents that traditional matched-model price indices—such as the Jevons index—often record steeper price declines than quality-adjusted alternatives in markets characterized by rapid model turnover and deep promotional cycles. Because matched-model indices fully incorporate temporary discounting and clearance pricing but do not adjust for quality improvements embedded in newly introduced models, they may overstate deflation in such settings.

Related but conceptually distinct evidence is provided by Ehrlich et al. (2023), who compare hedonic and exact demand-based price indices in settings with high product turnover. Rather than attributing discrepancies to any single index formula, they show that divergences between matched-model and quality-adjusted indices arise from differences in how product entry and exit, substitution patterns, and evolving product quality are handled at scale. Smartphones, with frequent model replacement, intensive discounting, and rapid specification upgrades, therefore constitute a natural environment in which traditional and hedonic measures of price change may diverge substantially.

Understanding how specific product features contribute to price formation—and how their influence evolves over time—is therefore central both to accurate inflation measurement and to industry pricing strategy. A substantial body of research emphasizes the importance of feature-based valuation in differentiated product markets (Pakes, 2003; Erickson and Pakes, 2011), documenting that attributes such as RAM, storage capacity, camera quality, and processor performance influence consumer willingness to pay. As technological progress standardizes once-premium features, their marginal contribution to price tends to decline, while newly introduced capabilities become key sources of product differentiation.

While traditional hedonic regressions provide a framework for quality adjustment, they face scalability challenges in high-dimensional and rapidly evolving markets (Ehrlich et al., 2023). Recent work shows that machine learning approaches can help address these limitations by enabling flexible, high-dimensional hedonic modeling with automatic feature selection and regularization (Tibshirani, 1996; Cafarella et al., 2023).

This paper applies a LASSO-based hedonic framework to smartphone models released between 2018 and 2025. Using longitudinal price data scraped from Amazon via the Keepa

API alongside detailed device specifications, the study estimates how the valuation of key features evolves over time and constructs hedonic Jevons indices that account for continuous improvements in product quality. By comparing these indices to traditional Jevons measures, the paper provides empirical evidence on how quality adjustment affects measured deflation in high-technology markets and contributes to ongoing discussions on inflation measurement in rapidly evolving product categories.

2 Literature Review

The pricing dynamics of mobile phones have been widely studied at the intersection of technological change, consumer behavior, and hedonic price measurement. Existing research emphasizes how rapid innovation, frequent product replacement, and evolving consumer preferences jointly shape observed price trajectories in consumer electronics markets.

A substantial literature documents the role of product characteristics in determining consumer valuation and pricing outcomes. The hedonic pricing framework introduced by Rosen (1974) formalizes how product attributes are implicitly priced in competitive markets. Empirical work on differentiated product demand further demonstrates that observable characteristics play a central role in shaping consumer willingness to pay and market pricing patterns (Berry et al., 1995). In technology markets, where product innovation is rapid and models are frequently replaced, incorporating detailed product specifications is particularly important for measuring price change and quality adjustment (Pakes, 2003; Aizcorbe et al., 2020). Together, this literature underscores the necessity of incorporating detailed product characteristics into pricing models.

Complementary work examines how innovation affects pricing strategies over the product lifecycle. Research on technology markets shows that rapid innovation leads to short product lifecycles and declining prices as new product generations enter and replace existing models (Norton and Bass, 1987). This pattern is consistent with earlier findings by Bayus (1991), who documents that technological obsolescence and feature improvements are central drivers of durable goods replacement behavior.

Hedonic Pricing Models and Quality Adjustment

Hedonic pricing models have long been used to adjust prices for changes in product quality. Seminal work by Pakes (2003) argues that conventional matched-model approaches can mismeasure price change when products are frequently replaced by higher-quality successors, emphasizing the need for index methods that explicitly account for innovation and product turnover. Extending this framework, Erickson and Pakes (2011) demonstrate how alternative index constructions can improve inflation measurement in dynamic product markets by incorporating richer information on product entry, exit, and quality change.

Ehrlich et al. (2023) further discusses the conceptual and empirical challenges of quality adjustment, particularly in settings with high-dimensional characteristics and rapid innovation. These concerns align with earlier work emphasizing that accurate price measurement must account for technological progress, product replacement, and evolving product characteristics over the lifecycle (Pakes, 2003; Erickson and Pakes, 2011; Bayus, 1991).

Recent research has directly examined systematic differences between traditional matched-model indices—such as the Jevons index—and hedonic indices in high-technology categories. The Bureau of Labor Statistics (2024) documents that in markets with rapid model turnover, deep promotional cycles, and substantial quality improvement—including televisions, wireless devices, and smartphones—matched-model indices frequently record steeper price declines than quality-adjusted alternatives. This divergence arises because matched-model methods fully incorporate temporary discounting and clearance pricing while failing to adjust for quality improvements embodied in newly introduced models.

Related evidence is provided by Ehrlich et al. (2023), who examine quality adjustment at scale by comparing hedonic and exact demand-based price indices in environments with high product turnover. Rather than attributing differences in measured price change to any single superior index formula, they show that divergences between matched-model and quality-adjusted indices arise from differences in how product entry and exit, substitution patterns, and evolving quality are treated in practice. In product categories characterized by frequent model replacement and intensive discounting, these methodological choices can materially affect measured price dynamics.

Taken together, this literature suggests that in markets such as smartphones—where rapid feature improvements (e.g., processor upgrades, camera modules, and memory expansion) coincide with intensive promotional activity and frequent model introductions—traditional Jevons indices are more likely to record steeper deflation than hedonic indices that explicitly control for evolving product quality.

Machine Learning Approaches in Hedonic Modeling

A key limitation of conventional hedonic regressions is scalability. As the number of product characteristics increases, traditional linear models face challenges related to overfitting and interpretability. Tibshirani (1996) introduces the Least Absolute Shrinkage and Selection Operator (LASSO) as a solution for high-dimensional settings, providing regularization and automatic variable selection. Building on this framework, Cafarella et al. (2023) demonstrate how regularized machine learning models can be used to construct hedonic price indices in markets characterized by rapid technological change and frequent product turnover.

Recent work has further expanded the role of machine learning in price measurement by leveraging large-scale electronic product data. Bajari et al. (2025) propose an AI-powered framework for constructing hedonic price indices in which machine learning models automatically extract product characteristics from high-dimensional sources such as product descriptions, images, and metadata. This approach substantially reduces the need for manually curated product attributes and enables the construction of quality-adjusted price indices at scale. Their results demonstrate that machine learning–based hedonic models can accurately recover constant-quality price movements even in markets characterized by rapid product turnover and complex product differentiation.

In the smartphone market, pricing dynamics are closely tied to the standardization of once-premium features and the introduction of new innovations. For example, the diffusion of connectivity technologies—from 3G to 4G/LTE and now to 5G—illustrates how the price contribution of a feature declines as it becomes ubiquitous. At the same time, persistent differences across operating systems, particularly between iOS and Android devices, continue to shape market segmentation and pricing patterns (Aizcorbe et al., 2020).

Overall, the existing literature provides strong theoretical and empirical foundations for analyzing smartphone prices using feature-based hedonic methods. Recent evidence from Bureau of Labor Statistics (2024) documents that in high-technology markets with rapid model turnover and deep promotional cycles, traditional matched-model indices often record steeper price declines than quality-adjusted alternatives. Complementary insights are provided by Ehrlich et al. (2023), who show that divergences between matched-model and hedonic indices arise from differences in the treatment of product entry and exit, temporary discounting, and other unobserved quality.

While some superlative index formulas may provide theoretically desirable properties, their performance in practice depends on the structure of the data and the characteristics of the market being studied. In markets with rapid product turnover and evolving product quality, different index constructions may capture different aspects of price dynamics. These considerations motivate the present study's use of LASSO-based hedonic models combined with longitudinal price data to more accurately measure constant-quality price change in the smartphone market.

3 Data

This study combines publicly available smartphone specifications with web-scraped historical price data to construct a panel dataset of mobile phone models. The base dataset is sourced from a Kaggle collection containing 931 smartphone models released between 2016 and 2025.[1] The dataset includes core hardware attributes such as RAM, battery capacity, screen size, processor type, and camera specifications, providing a comprehensive cross-sectional snapshot of mobile phone characteristics.

[1] https://www.kaggle.com/datasets/abdulmalik1518/mobiles-dataset-2025

3.1 Integrating Model Identifiers and Historical Prices

To obtain longitudinal pricing information, a custom web crawler was used to identify the Amazon Standard Identification Number (ASIN) corresponding to each phone model across North America Amazon listings. Because many models in the original Kaggle dataset were region-specific and not officially sold in North America, the ASIN-matching step resulted in 152 successfully identified models.

These ASINs were then queried through the Keepa API to retrieve daily historical prices. Since Keepa reports North America Amazon prices in USD by default, no currency conversion was required. To reduce noise and enhance comparability across time, quarterly average prices were computed for each model, producing a panel dataset suitable for analyzing both cross-sectional feature effects and intertemporal price trends.

3.2 Data Cleaning and Preprocessing

Several preprocessing steps were applied to harmonize the specification data and prepare the dataset for hedonic modeling:

- **Dropping Non-Predictive Identifiers:** Device name and model number were removed as they do not contain predictive information.
- **Categorical Encoding:** Processor types were mapped into three market-representative performance tiers—*entry-level*, *midrange*, and *flagship*—based on widely used benchmark groupings. Categorical features such as brand and processor tier were encoded using one-hot encoding, a procedure that converts each category into a binary indicator variable (taking the value 1 if the observation belongs to the category and 0 otherwise). This transformation allows categorical characteristics to enter the regression without imposing an artificial numerical ordering across categories.
- **Camera Feature Engineering:** The original rear camera specification was decomposed into two interpretable measures: *Max_MP* (the maximum megapixel value among rear lenses) and *Num_Back_Cameras* (the count of rear modules). This reflects the industry trend toward multi-camera systems and allows the model to isolate the contribution of camera quality versus camera quantity.
- **Missing Values:** Missing entries were imputed using the median of each feature to avoid distortion from extreme values.
- **Normalization:** All continuous features were scaled using the MinMaxScaler, which linearly rescales each variable to lie between 0 and 1 based on its observed minimum and maximum values. Normalization ensures that variables measured on different scales (e.g., battery capacity versus screen size) contribute proportionally during model estimation and prevents features with large numeric ranges from disproportionately influencing coefficient shrinkage in regularized regressions such as LASSO.

3.3 Final Dataset

After filtering models without valid ASINs and merging quarterly Keepa price histories, the final dataset consists of 152 smartphone models with longitudinal price observations and detailed technical specifications. This refined panel dataset enables the construction of both traditional and hedonic price indices and supports year-specific hedonic regressions to study how feature valuations evolve over time.

Overall, this integrated dataset provides a robust foundation for analyzing price dynamics in the smartphone market, capturing both cross-sectional hardware differences and temporal pricing behavior.

4 Methodology

The primary objective of this study is to identify how product characteristics influence smartphone prices over time and to evaluate how alternative econometric specifications affect the measurement of constant-quality price change. Because smartphone markets are characterized by rapid technological progress and frequent product turnover, isolating the contribution of individual features requires a modeling framework capable of handling high-dimensional predictors while maintaining interpretability. Cafarella et al. (2023) highlight the usefulness of sparsity-inducing methods for constructing hedonic price indices in such environments. Consistent with this approach, this study employs LASSO regression to identify the features most strongly associated with smartphone pricing.

The Least Absolute Shrinkage and Selection Operator (LASSO) is a regularized regression method that imposes an L1 penalty on coefficient magnitudes, shrinking less informative predictors toward zero and thereby producing a parsimonious specification. This property is particularly valuable in hedonic settings where many product attributes may be correlated.

4.1 Level-Based Hedonic Specification

Prices are modeled in logarithms, a standard transformation in hedonic price analysis that stabilizes variance and allows coefficients to be interpreted approximately as percentage effects. The baseline specification is:

$$\log(P_{it}) = \beta_0 + \sum_{j=1}^{p} \beta_j X_{ijt} + \epsilon_{it} \tag{1}$$

where:

- P_{it} denotes the observed price of smartphone i in period t,
- X_{ijt} represents the j^{th} product characteristic of device i in period t,
- β_j captures the marginal contribution of each feature to price,
- ϵ_{it} is an idiosyncratic error term.

For each period t, the LASSO estimator minimizes the following objective function:

$$\mathcal{L}_t(\beta_t) = \sum_{i=1}^{n_t} \left(\log(P_{it}) - \beta_{0t} - \sum_{j=1}^{p} \beta_{jt} X_{ijt} \right)^2 + \lambda_t \sum_{j=1}^{p} |\beta_{jt}| \tag{2}$$

where λ governs the strength of regularization. Larger values of λ produce sparser models by shrinking the coefficients on less important covariates toward zero.

Level-based regressions estimate the relationship between product characteristics and price in each period and generate predicted constant-quality prices that serve as inputs for hedonic price indices.

4.2 Price-Change Hedonic Specification

While level-based models are widely used in hedonic index construction, they rely on imputed price levels for non-overlapping products and may smooth short-run price movements. To evaluate the sensitivity of measured inflation to modeling strategy, this study also estimates a Price-Change specification that directly models price changes.

For smartphones observed in adjacent periods t and $t+1$, the dependent variable is defined as the log price difference:

$$\Delta \log(P_{it}) = \log(P_{i,t+1}) - \log(P_{it}) \tag{3}$$

The corresponding regression is:

$$\Delta \log(P_{it}) = \alpha_0 + \sum_{j=1}^{p} \alpha_j X_{ijt} + u_{it} \tag{4}$$

where:

- $\Delta \log(P_{it})$ denotes the log price change of smartphone i between periods t and $t+1$,

- X_{ijt} represents the j^{th} product characteristic of device i in period t,
- α_j captures the contribution of each feature to price changes,
- u_{it} is an error term capturing unexplained variation in price changes.

To address the high dimensionality of product characteristics, the model is estimated using LASSO regularization. For each adjacent period pair $(t, t+1)$, the estimator minimizes the following objective function:

$$\mathcal{L}_t(\alpha_t) = \sum_{i=1}^{n_t} \left(\Delta \log(P_{it}) - \alpha_{0t} - \sum_{j=1}^{p} \alpha_{jt} X_{ijt} \right)^2 + \lambda_t \sum_{j=1}^{p} |\alpha_{jt}| \tag{5}$$

where λ controls the strength of regularization. Larger values of λ encourage sparser models by shrinking less informative coefficients toward zero, allowing the model to select the most relevant product characteristics for explaining price changes.

This formulation estimates quality-adjusted price changes rather than price levels, anchoring predictions more closely to observed market movements and reducing the scope for imputation-driven divergence. Because the dependent variable is already differenced, the price-change approach mitigates concerns related to nonstationarity and limits the accumulation of prediction errors across periods. It therefore provides a useful complement to level-based hedonic models when constructing price indices in markets with rapid product turnover.

4.3 Application in this Paper

This study applies LASSO-based hedonic methods to analyze smartphone price dynamics and to construct quality-adjusted price indices under alternative modeling assumptions. Given the rapid pace of product turnover and feature innovation in the smartphone market, a single hedonic specification may not fully capture the range of mechanisms through which prices evolve over time. Accordingly, this paper implements a structured set of complementary models that differ along two key dimensions: (i) whether prices are modeled in levels or in changes, and (ii) whether temporal dependence is explicitly incorporated.

Level Hedonic Regression. As a starting point, the analysis estimates period-specific hedonic regressions in which the logarithm of price is modeled as a function of observable device characteristics. Separate LASSO regressions are estimated for each quarter (and, in parallel, for each year), allowing the implicit valuation of features such as RAM, camera quality, processor tier, and screen size to vary flexibly over time. By shrinking less informative coefficients toward zero, LASSO produces parsimonious specifications that isolate the most relevant attributes in each period while remaining robust in a high-dimensional setting.

Level Hedonic Regression with Lagged Error. To account for potential persistence in pricing dynamics, the baseline hedonic specification is augmented with a lagged prediction error as an additional regressor. In these sequential models, the hedonic regression for period t incorporates the prediction error from period $t-1$, enabling the model to learn from systematic deviations between observed and predicted prices. This approach captures time-series dependence that may arise from gradual demand shifts, pricing inertia, or unobserved market conditions.

Price-Change Hedonic Regression. In addition to modeling price levels, the study implements price-change specifications that directly model changes in log prices between adjacent periods. Rather than predicting prices independently in each period, these models estimate the relationship between product characteristics and price changes, thereby reducing the accumulation of level prediction errors over time. To ensure unbiased inference, price-change models are estimated using out-of-fold predictions generated through nested cross-validation, which prevents information from the target period from entering the estimation stage.

Price-Change Hedonic Regression with Lagged Error. The price-change framework is further extended by incorporating lagged prediction errors, allowing price changes to depend not only on contemporaneous characteristics but also on prior-period deviations. This dynamic structure is motivated by evidence that the relationship between prices and product characteristics evolves as market conditions shift, particularly in industries with frequent product entry and exit. Erickson and Pakes (2011) show that hedonic predictions must be updated across periods because valuations of characteristics change over time, and failure to account for these dynamics can distort measured price indices. By allowing past prediction errors to inform current estimates, the model captures these adjustment processes and improves the ability to track constant-quality price movements in rapidly evolving markets.

Pooled Time-Dummy Regression. As an alternative hedonic approach, the paper also estimates pooled regressions that combine observations from two adjacent periods and include a time indicator variable. In these models, a single set of feature coefficients is estimated jointly across periods, while time-dummy coefficients capture the average constant-quality price change between them. This approach has been widely used in the construction of quality-adjusted price indices for high-technology goods (Aizcorbe et al., 2020).

Formally, the pooled time-dummy specification can be written as:

$$\log(P_{it}) = \gamma_0 + \sum_{j=1}^{p} \gamma_j X_{ijt} + \delta_t + \varepsilon_{it} \tag{6}$$

where:

- P_{it} denotes the observed price of smartphone i in period t,
- X_{ijt} represents the j^{th} product characteristic,
- γ_j captures the implicit price of each product feature,
- δ_t is the time-dummy coefficient representing the average constant-quality price level in period t,
- ε_{it} is an error term.

In this framework, quality-adjusted price change between adjacent periods is obtained directly from the difference in the estimated time-dummy coefficients. By pooling observations across periods, the model improves estimation efficiency while maintaining a clear interpretation of the time effect as the constant-quality price shift between periods. Pooling also helps address price resets that occur when new product models replace older ones, a common feature of high-technology markets such as smartphones.

Price-Change Hedonic Regression with OLS. To assess whether the results are driven by the regularization inherent in LASSO, the price-change specification is re-estimated using ordinary least squares (OLS). These models retain the same dependent variable—log price differences between adjacent periods—and identical predictors, allowing the analysis to hold the functional form constant while relaxing the sparsity constraint. This comparison isolates the role of penalization in shaping measured price dynamics and provides a transparent benchmark for evaluating the stability of quality-adjusted indices. Consistency between LASSO and OLS estimates would indicate that the findings reflect underlying pricing behavior rather than artifacts of coefficient shrinkage.

Role in Price Index Construction. Across all specifications, predicted constant-quality prices (or predicted price changes) are used as inputs for constructing hedonic Jevons indices. By comparing indices derived from alternative modeling strategies—level-based, error-augmented, price-change-based, and time-dummy approaches—the analysis assesses the robustness of quality-adjusted price measurement to different assumptions about temporal dynamics and model structure. Together, these methods provide a comprehensive framework for evaluating how feature evolution and product turnover shape measured price change in the smartphone market.

4.4 Hyperparameter Choice

The LASSO regularization parameter α is selected via cross-validation in a way that is consistent across periods and model variants.

Baseline selection (all non-Price-Change models). For each period-specific regression, we use `LassoCV` with K-fold cross-validation to choose α that minimizes the cross-validated mean squared error. We set

$$K = \min\{5,\ \lfloor n/2 \rfloor\},$$

which caps the number of folds at five while ensuring each fold has sufficient observations in smaller samples. Predictors are standardized prior to fitting, and we fix `random_state`= 42 and `max_iter`= 2000 for reproducibility.

Search grid. `LassoCV` constructs a logarithmic grid of 100 candidate values between the data-driven $\alpha_{\max}$ (the smallest value that shrinks all coefficients to zero) and approximately $\alpha_{\max}/1000$. The selected $\hat{\alpha}$ is the value that achieves the best average validation performance. As sample size grows over time, the chosen $\hat{\alpha}$ typically decreases, allowing a less penalized (richer) specification.

Nested CV for Price-Change models. When modeling price *changes* (the Price-Change specification), we use nested cross-validation to avoid look-ahead bias and to obtain out-of-fold (OOF) predictions: (i) an *outer* K-fold partition produces held-out folds used solely for evaluation and OOF construction; (ii) within each outer training fold, an *inner* `LassoCV` selects $\hat{\alpha}$; (iii) the model is then refit on the outer training fold with the chosen $\hat{\alpha}$ and used to predict the held-out fold. This procedure yields OOF predictions for every observation and ensures that index construction and model comparison are based on predictions that never use their own outcomes in tuning.

Consistency across periods and variants. All period-specific models (quarterly and annual; Basic and Price-Change; with/without error features) share the same preprocessing pipeline, cross-validation rule for K, and tuning strategy for α. This harmonization stabilizes estimates in small samples and facilitates like-for-like comparisons between traditional and hedonic price indices.

4.5 Price Index Construction: Traditional, Level-Hedonic, and Price-Change Hedonic Jevons Indices

Building on the regression framework, this study constructs three alternative price indices to evaluate how modeling choices influence measured price dynamics in the smartphone market: the *Traditional Jevons Index*, a *Level-Based Hedonic Jevons Index*, and a *Price-Change Hedonic Jevons Index*. Together, these indices provide a structured comparison between matched-model approaches and quality-adjusted methods under differing econometric assumptions.

The traditional Jevons index relies exclusively on directly matched products observed in adjacent periods and therefore does not explicitly control for quality change. When product turnover is rapid, this approach may misattribute price differences arising from specification improvements or model replacement to pure price movements. Empirical evidence shows that many goods follow systematic life-cycle pricing patterns, with prices typically declining as products age and exiting items frequently sold at discounts, complicating the measurement of inflation (Melser and Syed, 2016). However, the direction of such bias is theoretically ambiguous: newer models may embody higher quality at similar prices, but they may also deliver improved performance without proportional price increases. As a result, quality adjustment does not mechanically imply higher measured inflation or lower deflation; the net effect is ultimately an empirical question.

To address this issue, the level-based hedonic index imputes constant-quality prices using predicted log price levels from feature-based regressions, allowing comparisons across non-identical products. In parallel, the price-change hedonic index directly models log price differences, anchoring the index more tightly to observed price movements while still controlling for observable characteristics.

By examining these three indices jointly, the analysis isolates the extent to which measured smartphone price changes depend on product matching, quality adjustment, and the econometric structure used to model price evolution. This multi-index framework provides a more comprehensive basis for assessing inflation in markets characterized by rapid innovation and frequent model entry and exit.

4.5.1 Traditional Jevons Index

Because the analysis models prices in logarithms, the Jevons index is implemented in its equivalent log formulation. For any pair of adjacent periods t and $t+1$, the index can be expressed as the exponential of the average log price relative:

$$J_{t,t+1} = \exp\left(\frac{1}{|M_{t,t+1}|}\sum_{i \in M_{t,t+1}} (\log p_{i,t+1} - \log p_{i,t})\right) \tag{7}$$

This formulation is algebraically equivalent to the geometric mean of price ratios but is computationally convenient when working with log-transformed prices.

4.5.2 Hedonic Jevons Index

To account for quality change, the study constructs a hedonic Jevons index using predicted constant-quality log prices from hedonic regressions. Let $\widehat{\log p}_{i,t}$ denote the predicted log price of device i in period t. The hedonic index is defined as:

$$HJ_{t,t+1} = \exp\left(\frac{1}{|S_{t,t+1}|}\sum_{i\in S_{t,t+1}}\left(\widehat{\log p}_{i,t+1} - \widehat{\log p}_{i,t}\right)\right) \tag{8}$$

By relying on predicted log prices, this approach enables price comparisons even when identical models are not observed in both periods.

4.5.3 Price-Change Hedonic Jevons Index

For price-change specifications, the index is constructed directly from predicted log price differences rather than predicted price levels. Let $\widehat{\Delta \log p}_{i,t,t+1}$ denote the predicted log price change for device i between periods t and $t+1$. The index is then given by:

$$HJ^{\Delta}_{t,t+1} = \exp\left(\frac{1}{|S_{t,t+1}|}\sum_{i\in S_{t,t+1}}\widehat{\Delta \log p}_{i,t,t+1}\right) \tag{9}$$

Constructing the index from predicted price changes avoids the accumulation of level prediction errors and reduces the dependence on imputed price levels.

4.5.4 Interpretation for Mobile Phone Markets

Smartphone markets are characterized by rapid product turnover, mid-cycle promotional discounts, and frequent quality improvements. These features complicate the measurement of constant-quality price change and motivate the use of multiple index constructions.

The traditional Jevons index fully incorporates observed price movements for matched models, capturing temporary discounts such as holiday promotions and end-of-life clearance pricing. However, because it excludes unmatched products, it may fail to reflect the price recovery associated with the introduction of higher-quality successor models, potentially overstating deflation in markets with rapid innovation.

Level-based hedonic indices address this limitation by imputing constant-quality prices using predicted log price levels. By holding product characteristics fixed, these indices isolate pure price movements and smooth short-lived fluctuations arising from promotional activity or model exit. As a result, they provide a measure of inflation that reflects underlying technological progress rather than transitory market dynamics.

Price-change hedonic indices take a complementary approach by directly modeling log price differences between adjacent periods. Because these models remain anchored to observed price adjustments, they limit the scope for imputation-driven smoothing while still controlling for quality change. Consequently, they offer an intermediate perspective that balances sensitivity to realized price movements with the need for constant-quality comparison.

Taken together, the three indices provide a structured framework for evaluating how methodological choices shape measured inflation in high-technology markets. Differences between them help disentangle the roles of promotional pricing, product replacement, and quality adjustment, allowing this study to assess whether observed deflation primarily reflects genuine price declines or the evolving characteristics of successive smartphone generations.

5 Results

This section presents the empirical results from the quarterly and annual implementations of the traditional and hedonic Jevons indices. Across both time aggregations, the results reveal systematic differences between level-based hedonic models and price-change-based specifications, as well as clear contrasts between traditional matched-model indices and quality-adjusted alternatives.

5.1 Regression Tables

Table 1: Level Regression Summary: 2021

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.9952***	[0.6320, 1.5338]	0.0100	0.2300
Mobile Weight (grams) [SELECTED]	-0.4431***	[-0.8591, -0.1092]	0.0500	0.1913
RAM Memory (GB) [SELECTED]	0.4599***	[0.3095, 0.6864]	0.0100	0.0962
Front Camera (MP) [SELECTED]	0.0778	[-0.0314, 0.2536]	0.7270	0.0727
Max Back Camera MP [SELECTED]	0.1518	[-0.3110, 0.3027]	0.7527	0.1566
Number of Cameras [SELECTED]	0.2220***	[0.0905, 0.3741]	0.0100	0.0723
Processor Level (encoded)	0.0000	[-0.1380, 0.1732]	0.0000	0.0794
Battery Capacity (mAh) [SELECTED]	0.2585	[-0.1169, 0.6329]	0.6553	0.1913
Screen Size (inches) [SELECTED]	0.2333*	[0.0601, 0.5916]	0.1000	0.1356

Sample Size: 47 R^2 Score: 0.8737 Optimal Alpha: 0.002142 Features Selected: 8/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Model fit increases sharply in 2021 ($R^2 = 0.87$), and nearly all features are selected. iOS carries a very large and highly significant premium, while RAM, number of cameras, and screen size also show strong positive effects. The richer sample and greater product differentiation in 2021 allow the model to identify clearer implicit prices, indicating a more structured pricing hierarchy across platform and hardware characteristics.

Table 2: Level Regression Summary: 2024

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.5086***	[0.4092, 0.6712]	0.0100	0.0668
Mobile Weight (grams) [SELECTED]	0.0472	[-0.1156, 0.1592]	0.8283	0.0701
RAM Memory (GB) [SELECTED]	0.3917***	[0.2554, 0.5342]	0.0100	0.0711
Front Camera (MP) [SELECTED]	0.1433***	[0.0294, 0.2639]	0.0500	0.0598
Max Back Camera MP [SELECTED]	0.0568	[-0.0434, 0.2019]	0.7686	0.0626
Number of Cameras [SELECTED]	-0.0993***	[-0.1812, -0.0104]	0.0500	0.0436
Processor Level (encoded) [SELECTED]	-0.0341	[-0.1294, 0.0523]	0.8123	0.0464
Battery Capacity (mAh) [SELECTED]	0.0873	[-0.0375, 0.2599]	0.7064	0.0759
Screen Size (inches) [SELECTED]	0.0071	[-0.1222, 0.1663]	0.9755	0.0736

Sample Size: 142 R^2 Score: 0.5598 Optimal Alpha: 0.003693 Features Selected: 9/9
[SELECTED] = Feature selected by Lasso Significance: *** $|\text{coef}| > 2\cdot\text{SE}$, ** $|\text{coef}| > 1.96\cdot\text{SE}$, * $|\text{coef}| > 1.645\cdot\text{SE}$

The 2024 regression maintains moderate explanatory power ($R^2 = 0.56$). The iOS premium remains stable at roughly 0.5 log points, and RAM continues to exhibit a strong positive effect. Notably, the number of cameras becomes significantly negative, reinforcing evidence of feature saturation in imaging hardware. Incremental increases in traditional hardware specifications appear to yield declining marginal pricing power.

The contrast between 2021 and 2024 illustrates a clear shift in smartphone pricing dynamics. In 2021, multiple hardware attributes—including camera count, RAM, and screen size—command strong positive premia, indicating a highly differentiated product space. By 2024, however, the marginal value of additional camera modules becomes negative, suggesting that once-premium hardware features had become standardized, leaving platform and memory capacity as the primary drivers of price dispersion.

Table 3: Price-Change Regression Summary: 2020 → 2021

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.0236	[-0.0031, 0.1100]	0.7917	0.0289
Mobile Weight (grams)	0.0000	[-0.0170, 0.0183]	0.0000	0.0090
RAM Memory (GB) [SELECTED]	-0.0164	[-0.0789, 0.0243]	0.8407	0.0263
Front Camera (MP)	0.0000	[-0.0255, 0.0367]	0.0000	0.0159
Max Back Camera MP [SELECTED]	0.0349	[0.0000, 0.0991]	0.6479	0.0253
Number of Cameras [SELECTED]	-0.0287***	[-0.0588, -0.0111]	0.0500	0.0122
Processor Level (encoded)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Battery Capacity (mAh)	0.0000	[-0.0179, 0.0359]	0.0000	0.0137
Screen Size (inches)	0.0000	[0.0000, 0.0165]	0.0000	0.0042

Sample Size: 24 R^2 Score: 0.6106 Optimal Alpha: 0.005357 Features Selected: 4/9
[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

The price-change regression achieves substantial explanatory power ($R^2 = 0.61$), indicating that short-run price adjustments were partly linked to evolving hardware characteristics. The only statistically significant effect is a negative coefficient on the number of cameras, suggesting that additional cameras were associated with slightly faster price declines during this transition period.

Table 4: Price-Change Regression Summary: 2022 → 2023

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[-0.0122, 0.0000]	0.0000	0.0031
Mobile Weight (grams)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
RAM Memory (GB)	0.0000	[0.0000, 0.0499]	0.0000	0.0127
Front Camera (MP)	0.0000	[0.0000, 0.0659]	0.0000	0.0168
Max Back Camera MP	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Number of Cameras	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Processor Level (encoded)	0.0000	[-0.0238, 0.0000]	0.0000	0.0061
Battery Capacity (mAh)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Screen Size (inches)	0.0000	[0.0000, 0.0455]	0.0000	0.0116

Sample Size: 86 R^2 Score: 0.0000 Optimal Alpha: 0.037075 Features Selected: 0/9

The LASSO procedure selects no features and yields $R^2 = 0$, indicating that observable specification differences explain virtually none of the cross-sectional variation in price changes. This period appears dominated by common shocks or uniform repricing rather than feature-driven adjustment, highlighting the limits of differenced hedonic models in low-signal environments.

5.2 Coefficient Trend Interpretation

This section examines the time variation in selected characteristic coefficients estimated under the Basic Hedonic specification. Because the dependent variable is log price, coefficients are interpreted as semi-elasticities. Annual regressions provide smoother long-run valuation patterns, while quarterly estimates highlight short-run volatility.

RAM Capacity (`ram_mem_numeric`)

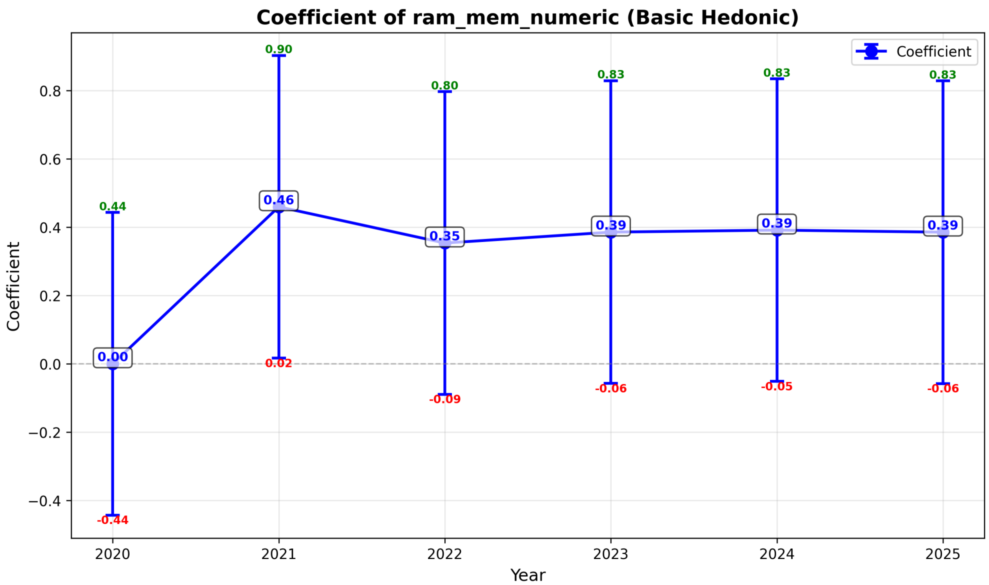

Figure 1: Annual coefficient of RAM capacity under the Basic Hedonic model.

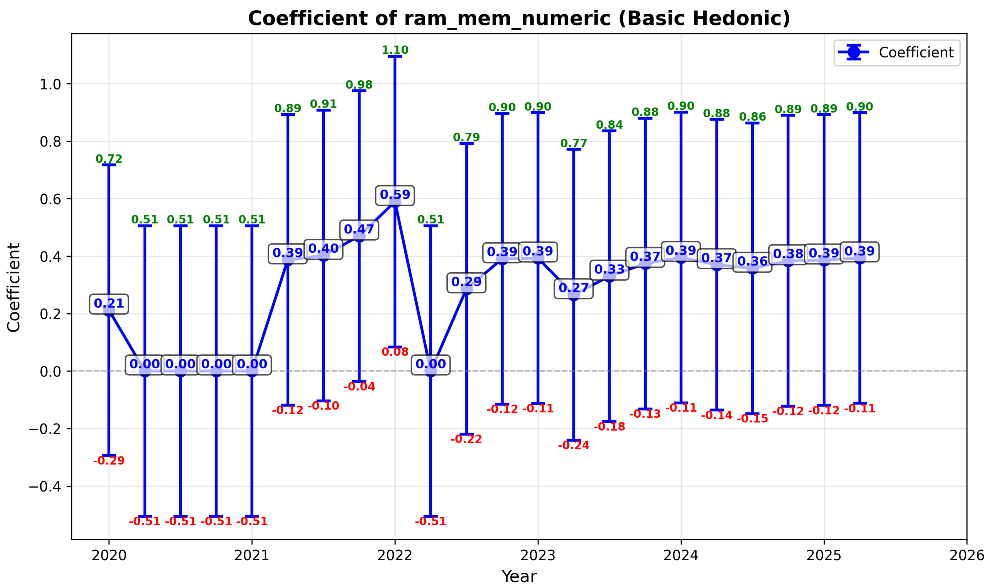

Figure 2: Quarterly coefficient of RAM capacity under the Basic Hedonic model.

The coefficient on RAM capacity is consistently positive and economically large. Annual estimates range roughly between 0.35 and 0.46, indicating a substantial price premium for higher memory configurations. Quarterly estimates show temporary spikes but remain positive in most periods. This persistence suggests that RAM continues to serve as a key vertical differentiation margin rather than a fully commoditized feature.

Number of Cameras (`num_cameras_numeric`)

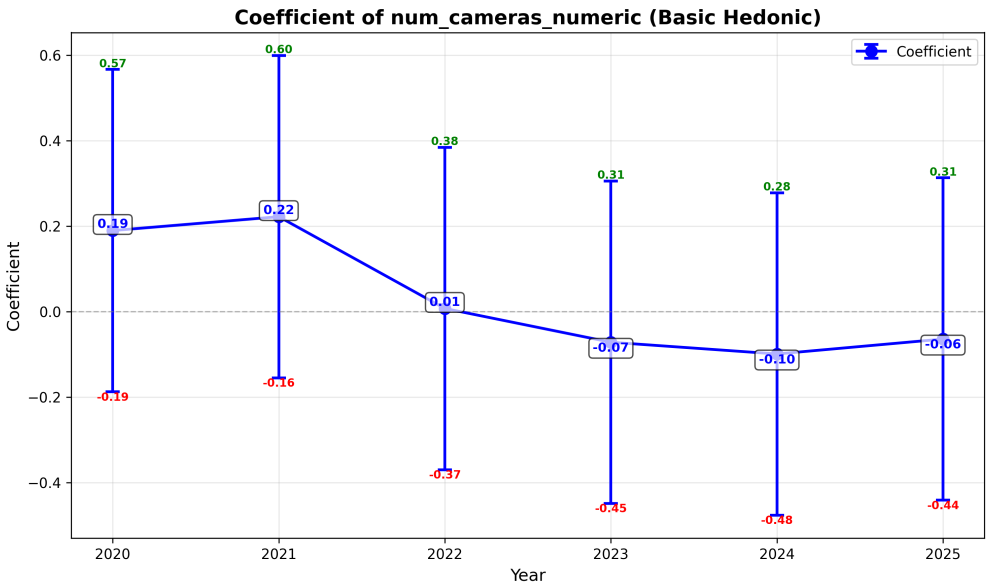

Figure 3: Annual coefficient of number of cameras under the Basic Hedonic model.

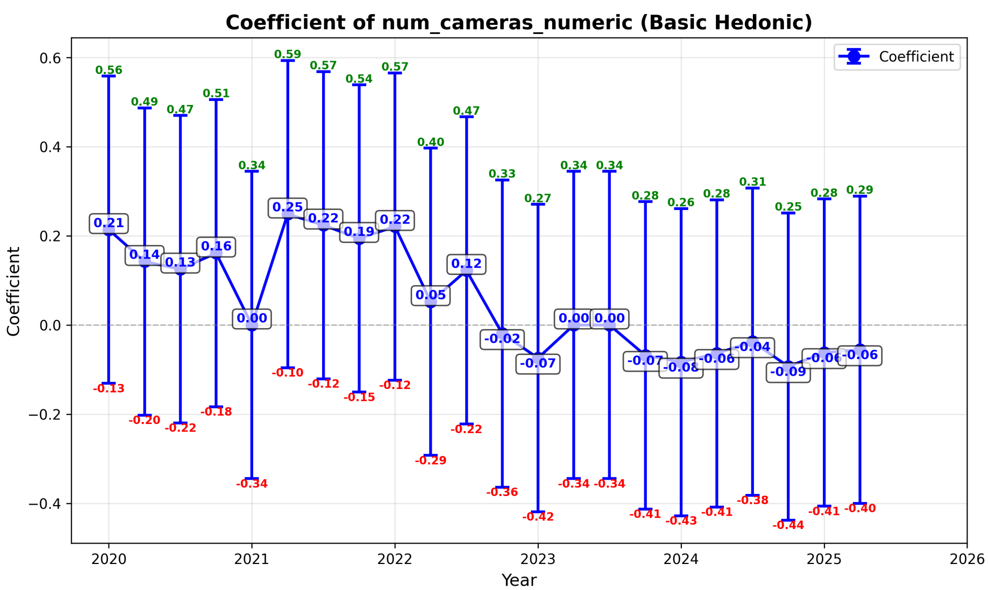

Figure 4: Quarterly coefficient of number of cameras under the Basic Hedonic model.

The number of cameras carries a positive coefficient in early periods, particularly during 2020–2021 when multi-camera systems expanded rapidly. However, the magnitude declines over time and becomes less stable in quarterly regressions. This pattern is consistent with feature normalization: once multi-camera setups become standard across tiers, additional cameras generate diminishing marginal price differentiation.

Maximum Megapixels (`max_mp_numeric`)

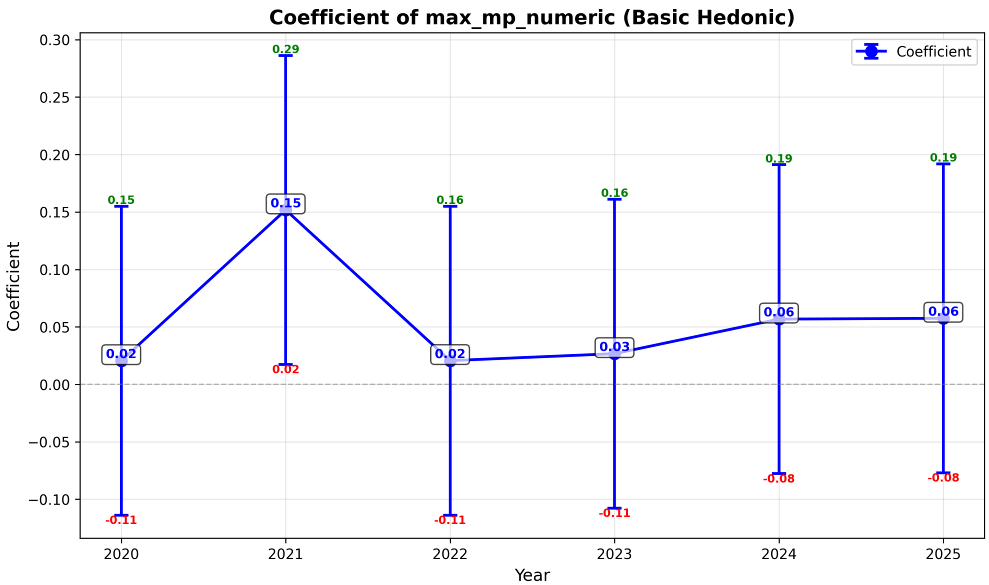

Figure 5: Annual coefficient of maximum megapixels under the Basic Hedonic model.

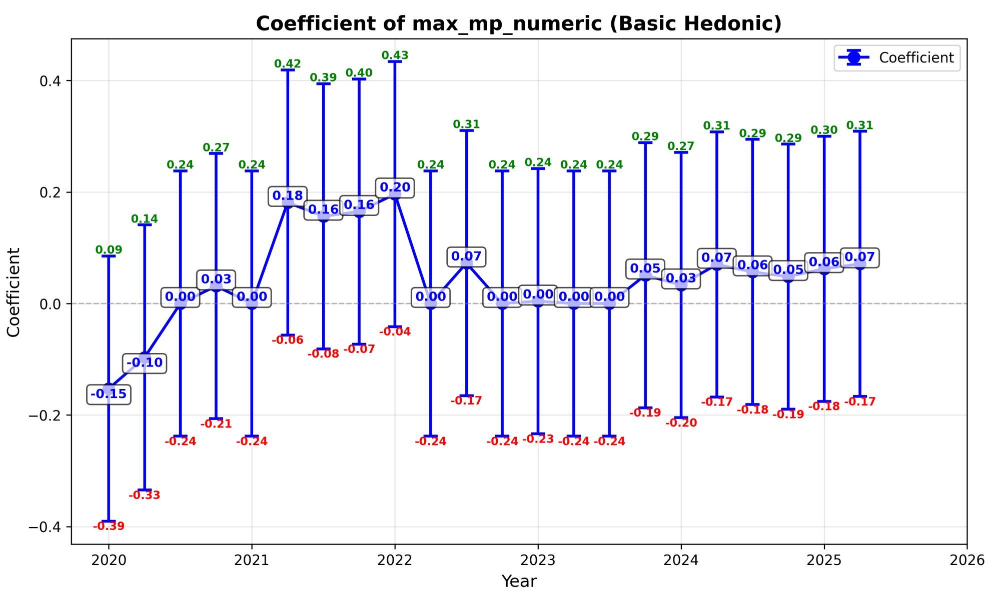

Figure 6: Quarterly coefficient of maximum megapixels under the Basic Hedonic model.

The coefficient on maximum megapixels is positive but exhibits considerable dispersion in quarterly estimates. Annual values remain modest in magnitude. The instability suggests that raw resolution has become a weaker standalone determinant of price, possibly reflecting the growing importance of computational photography and sensor quality over headline megapixel counts.

Device Weight (`mobile_weight_numeric`)

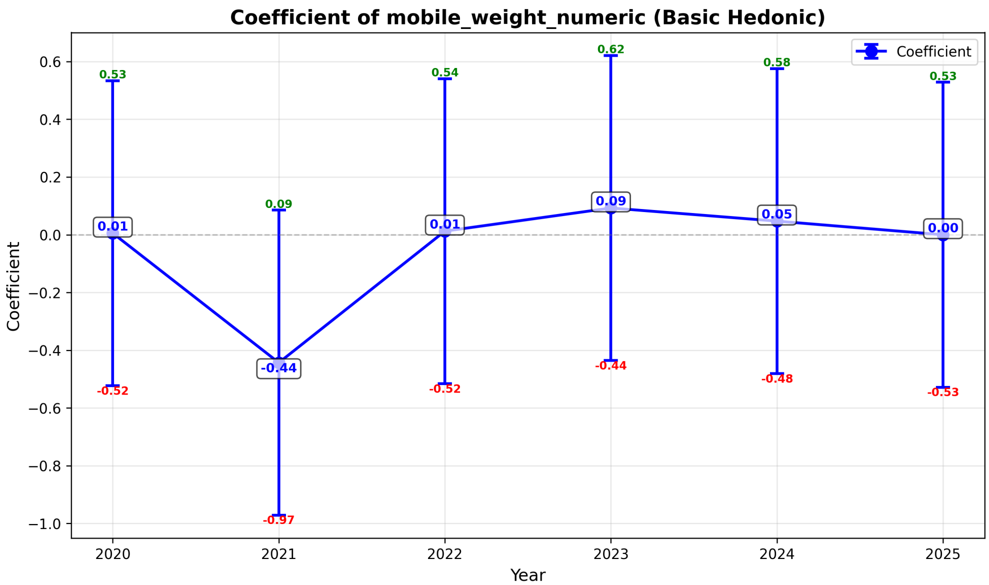

Figure 7: Annual coefficient of device weight under the Basic Hedonic model.

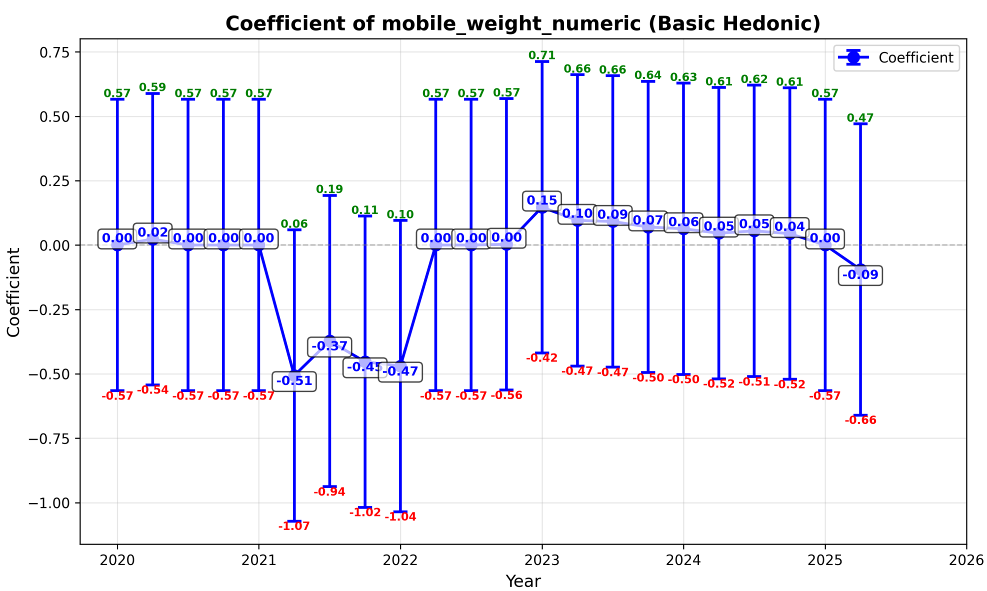

Figure 8: Quarterly coefficient of device weight under the Basic Hedonic model.

The coefficient on device weight fluctuates in sign and magnitude across specifications. In several periods the estimate is close to zero, and confidence intervals are wide. This suggests that weight does not exert a stable marginal valuation effect once correlated hardware characteristics are controlled for. Device weight appears to be a secondary attribute whose pricing relevance depends on broader design trade-offs.

Battery Capacity (`battery_capacity_numeric`)

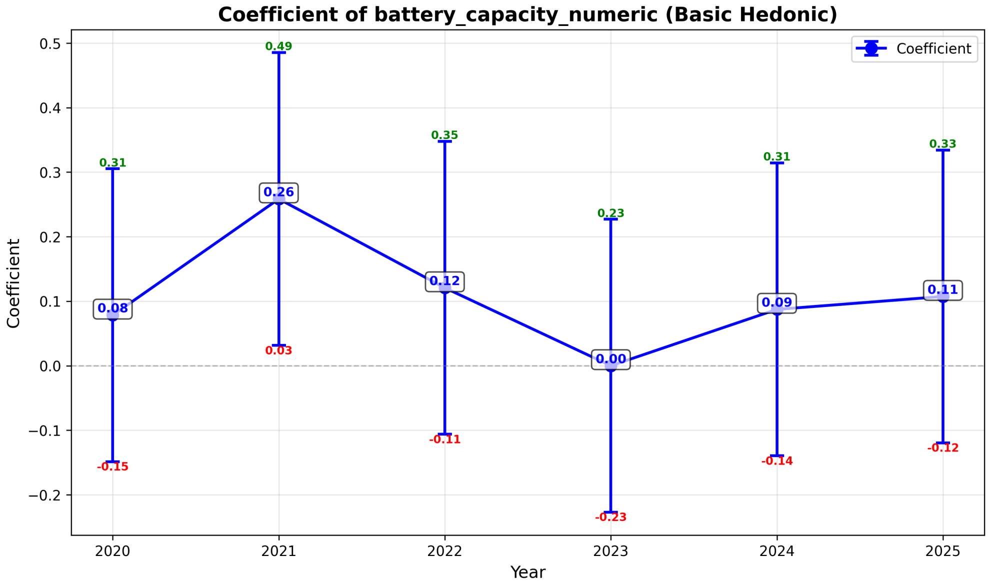

Figure 9: Annual coefficient of battery capacity under the Basic Hedonic model.

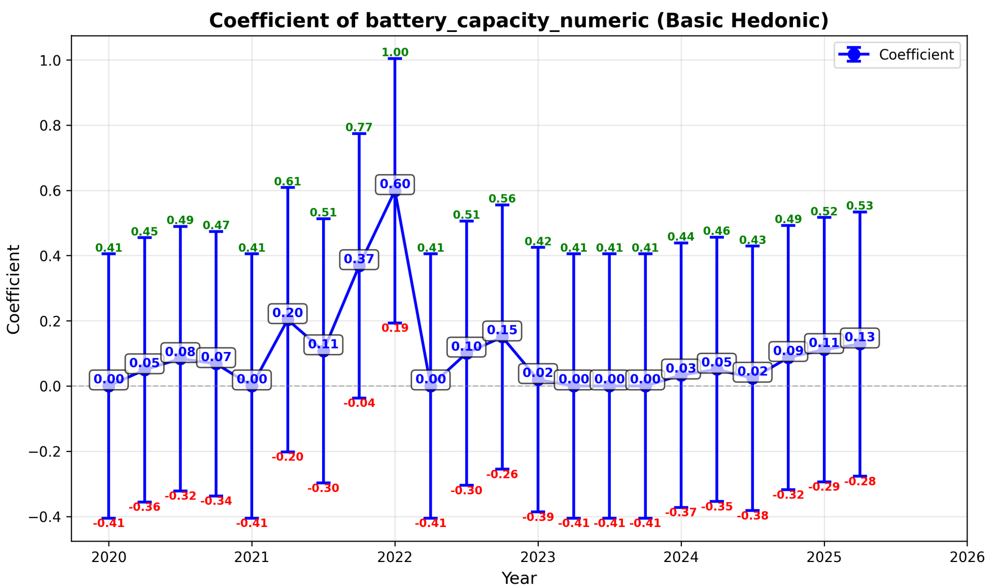

Figure 10: Quarterly coefficient of battery capacity under the Basic Hedonic model.

Battery capacity generally carries a positive coefficient in annual regressions, indicating a premium for larger batteries. However, quarterly estimates display periods of attenuation toward zero. This pattern is consistent with gradual standardization: as baseline battery sizes increase across models, incremental capacity improvements yield smaller marginal willingness to pay.

Operating System Indicator (`is_ios`)

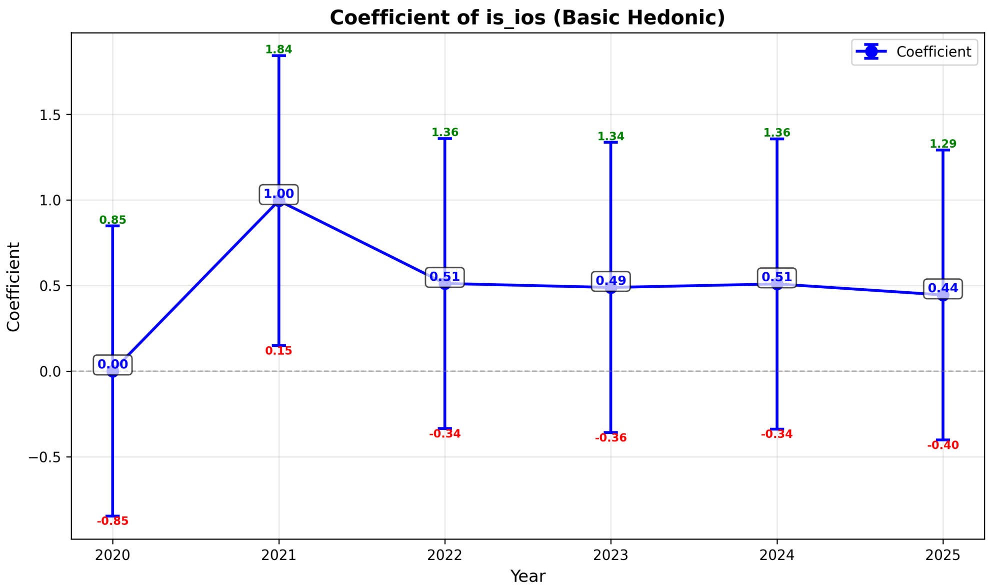

Figure 11: Annual coefficient of iOS indicator under the Basic Hedonic model.

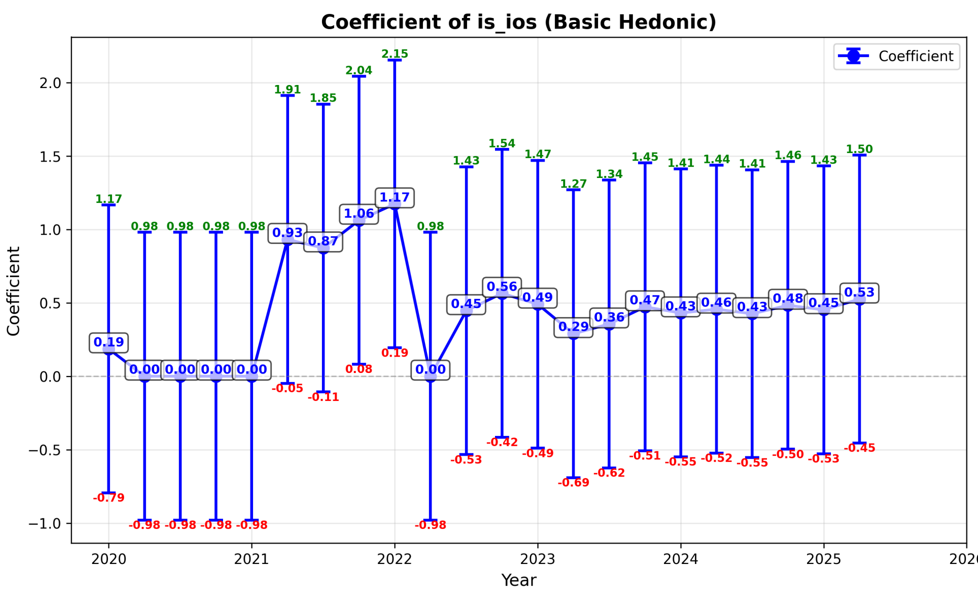

Figure 12: Quarterly coefficient of iOS indicator under the Basic Hedonic model.

The iOS indicator exhibits a consistently positive and economically large coefficient across all specifications. Annual estimates imply a substantial price premium for iOS devices relative to Android devices after controlling for observable hardware characteristics. Unlike many hardware features, the operating system premium remains persistent over time, suggesting that ecosystem integration and brand effects constitute durable components of smartphone pricing.

Overall, the coefficient trajectories reveal a structural contrast. Core performance attributes such as RAM maintain stable positive valuation effects. Several hardware features (camera count, megapixels, battery capacity) display signs of gradual commoditization, reflected in declining or unstable coefficients. In contrast, platform-level differentiation, captured by the iOS indicator, remains persistently strong, indicating that ecosystem effects dominate hardware-based differentiation in later periods.

5.3 Quarterly Indices Results

This subsection presents quarterly inflation dynamics and cumulative chained indices under traditional and hedonic approaches. Together, these results reveal how alternative modeling strategies shape measured price change in the smartphone market.

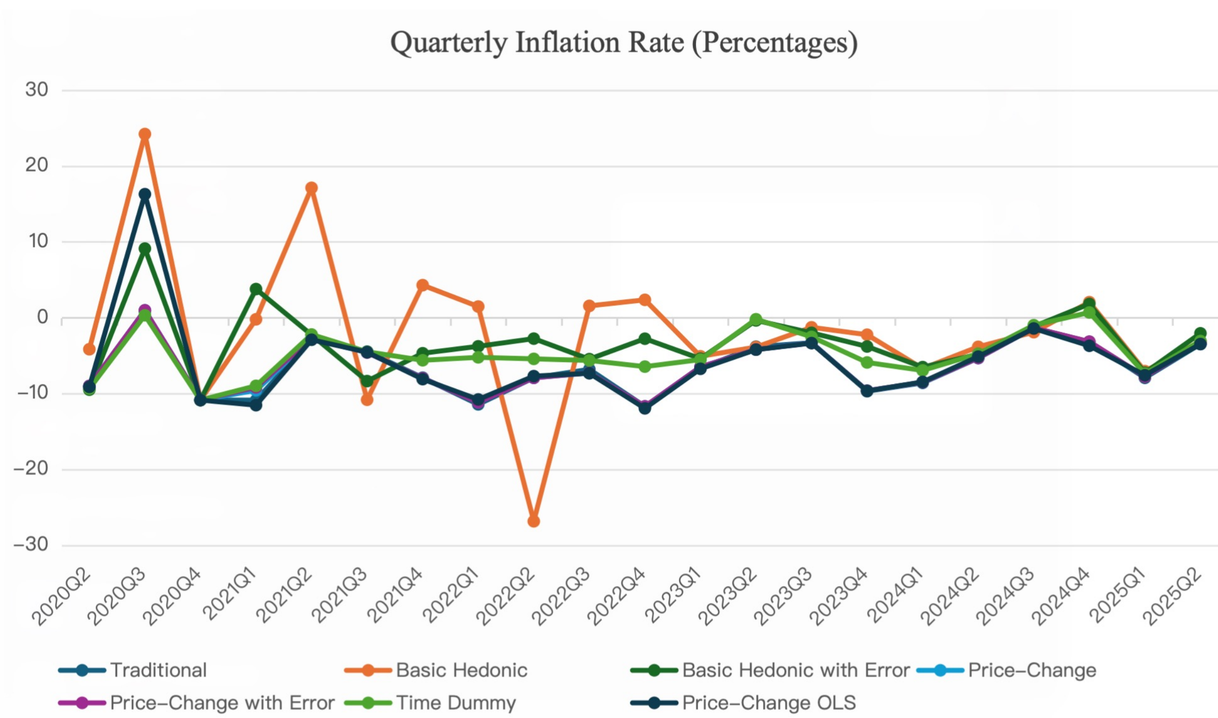

Figure 13: Quarterly inflation rates across traditional and hedonic index specifications.

Quarterly Inflation Rates. Figure 13 plots quarter-to-quarter inflation rates. The traditional matched-model index exhibits pronounced volatility, reflecting the influence of promotional discounting, inventory clearance, and rapid product replacement cycles.

Level-based hedonic models produce smoother inflation paths, indicating that controlling for evolving product characteristics dampens short-run price fluctuations. By contrast, price-change specifications closely track the traditional series, suggesting that once models are differenced, much of the volatility reflects genuine price adjustments rather than compositional changes in product quality.

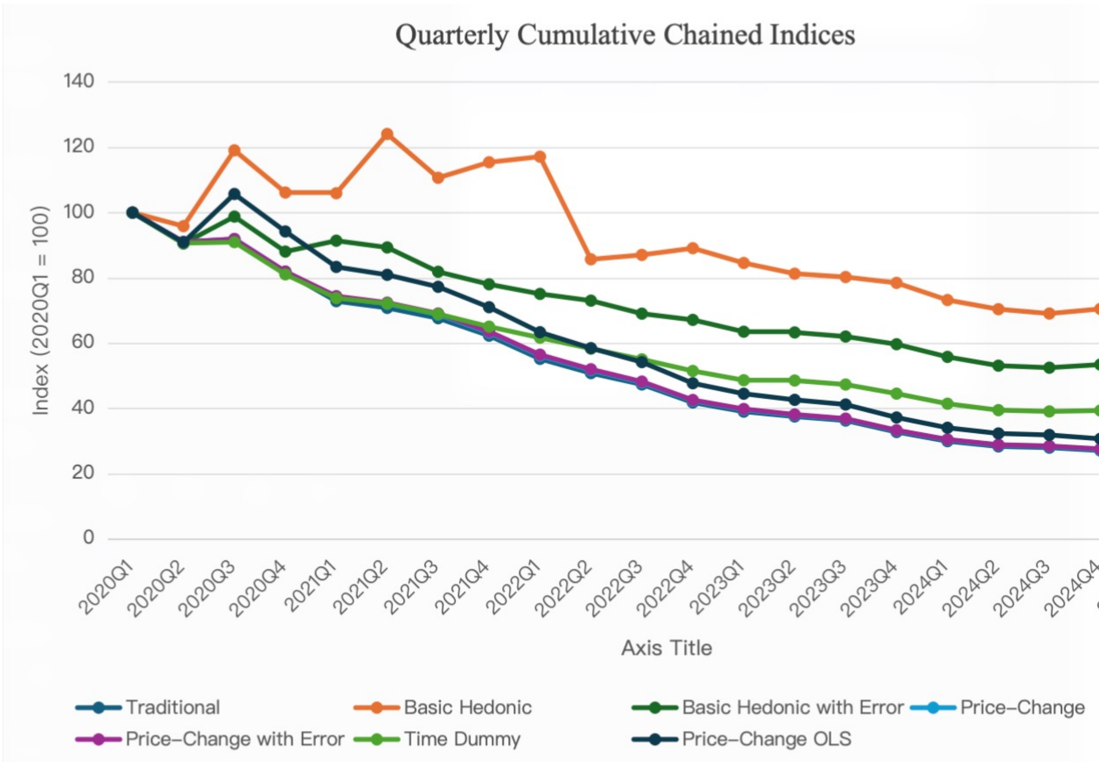

Figure 14: Quarterly cumulative chained indices under alternative modeling approaches.

Cumulative Chained Indices. Figure 14 reports cumulative chained indices beginning in 2020Q1. The traditional Jevons index indicates a cumulative price decline exceeding 71 percent, highlighting the combined effects of discounting and model turnover.

Level-based hedonic models imply substantially smaller declines, consistent with the interpretation that part of the observed deflation reflects quality improvement rather than pure price reductions. In contrast, price-change models generate trajectories that nearly coincide with the traditional index, suggesting limited incremental quality adjustment once prices are modeled in differences.

Model Comparison and Explanatory Power. Table 5 reports the final chained Jevons indices, implied cumulative deflation, and explanatory power measured in the price-change dimension for each specification.

Table 5: Quarterly cumulative deflation and R^2 (price-change dimension) across alternative index constructions.

Model	Final Chained Jevons Index	Cumulative Deflation	R^2 (price-change)
Traditional Jevons	-1.4231	-75.90%	N/A
Levels Hedonic	-0.6759	-49.13%	0.1279
Levels Hedonic with Lagged Errors	-0.7226	-51.45%	0.0936
Price-Change Hedonic	-1.4110	-75.61%	0.2559
Price-Change Hedonic with Lagged Errors	-1.4020	-75.39%	0.2772
Price-Change Hedonic with OLS	-1.2952	-72.62%	0.4364
Levels Hedonic with Time-Dummy Variables	-1.0399	-64.65%	0.0294 (partial R^2)

Several patterns emerge. First, the traditional matched-model index and the price-change hedonic specifications yield nearly identical cumulative declines. The traditional index implies a cumulative deflation of -75.90%, while the baseline price-change hedonic model records -75.61%, and the error-augmented price-change model yields -75.39%. This close alignment indicates that directly modeling log price differences largely preserves the high-frequency movements captured by the matched-model approach.

Second, level-based hedonic specifications imply substantially smaller cumulative deflation. The baseline levels hedonic model produces a decline of -49.13%, while incorporating lagged prediction errors modestly increases the decline to -51.45%. When coefficients are pooled across periods using time-dummy variables, the cumulative decline is -64.65%, lying between the pure levels and price-change estimates. These results suggest that modeling prices in levels, particularly with coefficient stability across time, leads to stronger quality adjustment and therefore less measured deflation.

Turning to explanatory power, R^2 is reported uniformly in the price-change dimension to ensure comparability across specifications. Price-change models achieve the highest predictive performance, with $R^2 = 0.2559$ for the baseline specification, 0.2772 when lagged errors are included, and 0.4364 under OLS estimation. The higher R^2 for the OLS price-change model suggests that removing LASSO regularization increases in-sample fit, although this may reflect mild overfitting in a noisy differenced setting.

By contrast, when level-based models are evaluated on their implied price changes (i.e., predicted log-price differences), explanatory power is substantially lower. The levels hedonic model achieves $R^2 = 0.1279$, and the lagged-error variant yields 0.0936. For the pooled time-dummy specification, explanatory power is measured using the *partial* R^2 of the time-dummy coefficients rather than the standard price-change regression R^2. Because time dummies enter the pooled regression as period-specific intercept shifts, the implied predicted price change between adjacent periods is identical for all products. Specifically, the model implies

$$\Delta \widehat{\log(P_{it})} = \hat{\delta}_{t+1} - \hat{\delta}_t,$$

which does not vary across i. As a result, the predicted price change has essentially no cross-sectional variation within each interval. When these nearly constant predictions are regressed on the observed cross-sectional price changes, the variance of the fitted values is close to zero, so the resulting R^2 mechanically approaches zero. This reflects the fact that time-dummy models capture the average price shift between periods rather than cross-sectional variation in price changes.

Instead, we compute the partial R^2 from the pooled levels regression, defined as

$$R^2_{\text{partial,time}} = 1 - \frac{\text{SSE}_{\text{full}}}{\text{SSE}_{\text{reduced}}},$$

where the full model includes both characteristics and time dummies, and the reduced model excludes the time dummies. The resulting partial $R^2 = 0.0821$ indicates that, conditional on product characteristics, time dummies explain approximately 8% of the remaining variation in log prices. This measure more accurately captures the economic contribution of the time-dummy structure to constant-quality price shifts.

Taken together, the quarterly evidence highlights two central insights. First, measured cumulative deflation is highly sensitive to whether prices are modeled in levels or in changes. Second, explanatory power in the price-change dimension aligns more closely with price-change specifications, suggesting that directly modeling log price differences better captures short-run pricing dynamics in the smartphone market.

5.4 Annual Indices Results

This subsection reports annual inflation dynamics and cumulative chained indices across traditional and hedonic specifications. Aggregating to the yearly level smooths short-term promotional volatility and provides a complementary perspective on constant-quality price change in the smartphone market.

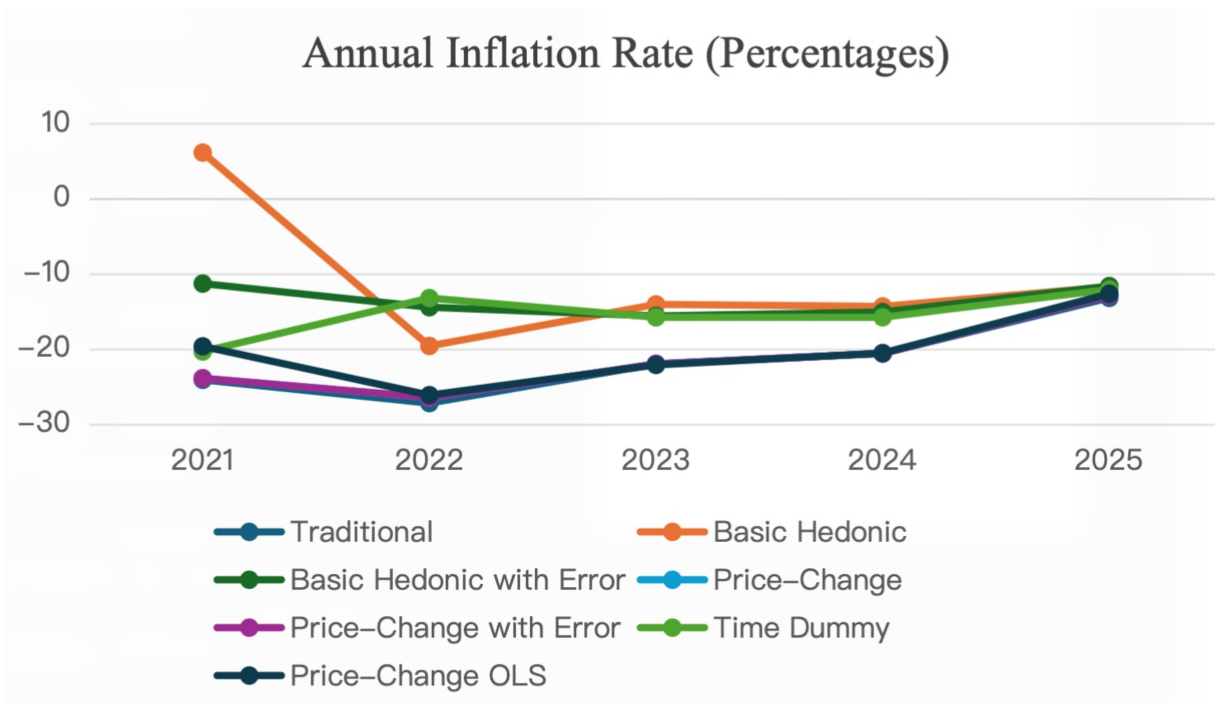

Figure 15: Annual inflation rates across traditional and hedonic index specifications.

Annual Inflation Rates. Figure 15 presents year-to-year inflation rates. Relative to quarterly estimates, inflation paths are noticeably smoother, reflecting the averaging out of temporary discounts and short-lived pricing shocks.

The traditional index continues to show persistent deflation throughout the sample period, although the magnitude varies across years. Level-based hedonic models again produce less extreme movements, indicating that a portion of observed price declines is attributable to quality improvement rather than pure price reductions. Price-change models, by contrast, closely track the traditional series, reinforcing the interpretation that differencing prices limits the scope for additional quality adjustment.

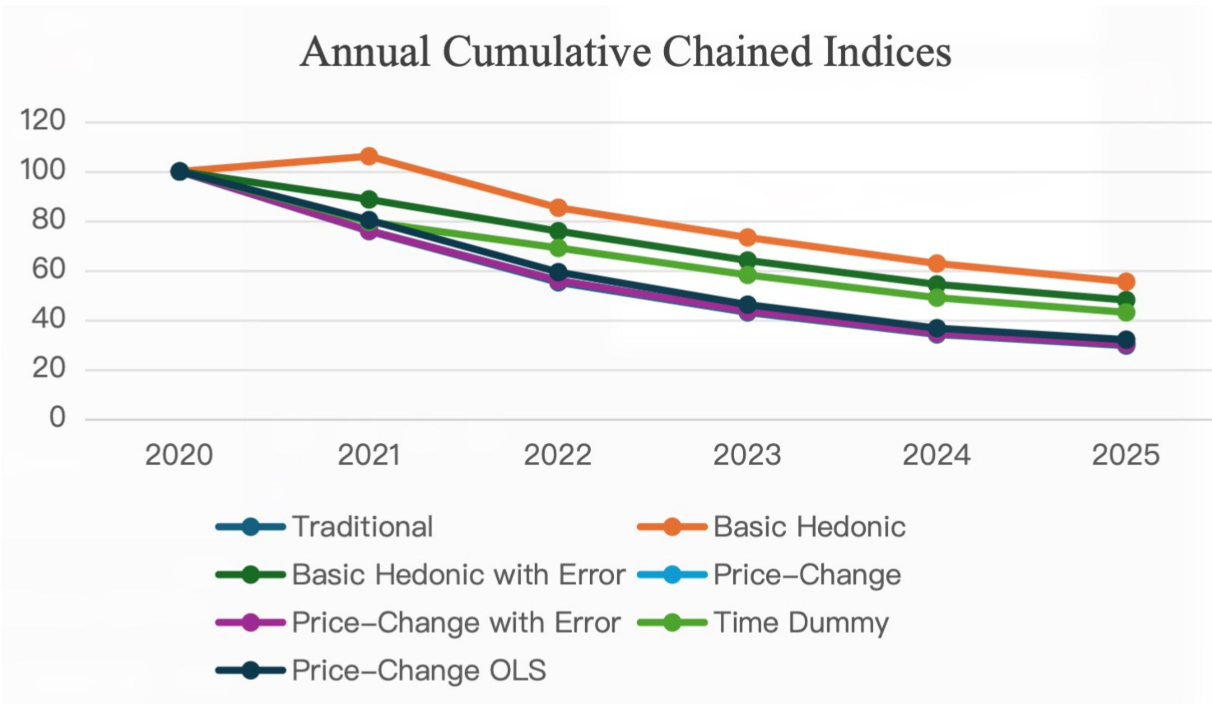

Figure 16: Annual cumulative chained indices under alternative modeling approaches.

Cumulative Chained Indices. Figure 16 reports chained indices beginning in 2020. The traditional annual Jevons index indicates a cumulative price decline of approximately 76 percent over the sample horizon.

Consistent with the quarterly results, level-based hedonic models imply substantially smaller declines. The basic hedonic specification records cumulative deflation of roughly −73.9 percent, while the error-augmented variant produces a nearly identical estimate. In contrast, price-change models generate trajectories that lie very close to the traditional index, suggesting limited incremental quality adjustment once prices are modeled in differences.

Model Comparison and Explanatory Power. Table 6 reports the final chained Jevons indices, implied cumulative deflation, and explanatory power measured in the price-change dimension for annual specifications.

Table 6: Annual cumulative deflation and R^2 (price-change dimension) across alternative index constructions.

Model	Final Chained Jevons Index	Cumulative Deflation	R^2 (price-change)
Traditional Jevons	-1.2135	-70.29%	N/A
Levels Hedonic	-0.7387	-52.22%	0.1915
Levels Hedonic with Lagged Errors	-0.7325	-51.93%	0.1012
Price-Change Hedonic	-1.1970	-69.79%	0.2498
Price-Change Hedonic with Lagged Errors	-1.1968	-69.79%	0.2623
Price-Change Hedonic with OLS	-1.1371	-67.92%	0.4226
Levels Hedonic with Time-Dummy Variables	-0.8412	-56.88%	0.0821 (partial R^2)

Several patterns mirror the quarterly results. First, the traditional matched-model index and the price-change hedonic models again produce nearly identical cumulative declines. The traditional index implies cumulative deflation of -70.29%, while the baseline price-change hedonic model yields -69.79%, and the lagged-error variant produces the same -69.79%. These close estimates confirm that modeling log price differences directly preserves the overall deflationary pattern observed in matched-model comparisons.

Second, level-based hedonic specifications imply materially less deflation. The baseline levels model yields -52.22%, and the lagged-error variant produces a nearly identical -51.93%. The pooled time-dummy specification generates an intermediate estimate of -56.88%, again suggesting that imposing coefficient stability across periods leads to stronger quality adjustment relative to price-change models.

Turning to explanatory power, R^2 is uniformly evaluated in the price-change dimension to ensure comparability. As in the quarterly case, price-change models exhibit the strongest predictive performance, with $R^2 = 0.2498$ for the baseline specification, 0.2623 when lagged errors are included, and 0.4226 under OLS estimation. The higher R^2 under OLS reflects greater in-sample fit once regularization is removed, though this may also indicate increased sensitivity to noise in annual price differences.

When level-based models are evaluated based on their implied price changes, explanatory power is lower. The levels specification achieves $R^2 = 0.1915$, and the lagged-error version yields 0.1012. For the pooled time-dummy specification, explanatory power is measured using the *partial* R^2 of the time-dummy coefficients rather than the standard price-change regression R^2.

Because time dummies enter the pooled levels regression as period-specific intercept shifts, their implied predicted price changes are nearly constant within each adjacent annual interval. Regressing these near-constant predictions on cross-sectional price changes mechanically yields an R^2 close to zero and understates the economic role of the time-dummy structure.

Instead, we compute the partial R^2 from the pooled levels regression by comparing the residual sum of squares of the full model (including characteristics and time dummies) with that of the reduced model (including characteristics only). The resulting partial $R^2 = 0.0821$ indicates that, conditional on product characteristics, time dummies explain approximately 8% of the remaining variation in log prices. This measure more appropriately captures the contribution of pooled time effects to constant-quality annual price shifts.

Overall, the annual evidence reinforces two central findings. First, cumulative deflation estimates remain highly sensitive to whether prices are modeled in levels or in changes. Second, explanatory power in the price-change dimension aligns more closely with price-change specifications, indicating that directly modeling log price differences better captures constant-quality price dynamics even at lower frequencies.

Summary of Patterns

Across both quarterly and annual implementations, three robust patterns emerge. First, the traditional matched-model Jevons index consistently records the steepest cumulative price declines. Second, level-based hedonic models imply materially smaller deflation, while price-change-based hedonic models generate cumulative price paths that closely track the traditional index despite explicitly controlling for product characteristics. This contrast highlights that modeling price differences directly preserves much of the high-frequency variation captured by matched-model approaches.

Third, modeling structure plays a decisive role in shaping measured quality adjustment. Specifications that impose additional structure—such as pooled time-dummy regressions or alternative regularization schemes—yield intermediate results and can alter cumulative deflation estimates by more than 15–20 percentage points. Notably, the explanatory power of time-dummy specifications is better captured by partial R^2 within the pooled levels regression, reflecting their role in shifting price levels rather than explaining cross-sectional price-change variation.

Taken together, these findings indicate that measured deflation in smartphone prices is highly sensitive to econometric design choices, including whether prices are modeled in levels or differences, how product entry and exit are treated, and how imputation is implemented. Importantly, quality adjustment does not mechanically imply higher or lower measured inflation; in principle, bias can arise in either direction depending on market dynamics and model structure. The implications of these differences for price measurement and interpretation are explored in the following discussion.

6 Conclusion

This paper evaluates how alternative index constructions measure price change in markets characterized by rapid product turnover, frequent discounting, and continuous quality improvement, using smartphones as a representative high-technology product category. Rather than presuming that hedonic methods mechanically produce smaller measured inflation, the analysis demonstrates that the magnitude — and even the direction — of divergence from traditional matched-model indices depends fundamentally on model structure.

Across both quarterly and annual frequencies, the traditional matched-model Jevons index records the steepest cumulative price declines. This result is consistent with recent evidence from Bureau of Labor Statistics (2024), reflecting the fact that matched-model indices fully incorporate temporary end-of-life discounts while omitting the price recovery associated with higher-quality successor models. In this setting, measured deflation is amplified because promotional price cuts are retained while quality improvements embodied in replacement products are excluded.

Quality-adjusted hedonic indices, however, do not move uniformly in the opposite direction. Level-based hedonic models imply substantially smaller cumulative deflation, particularly when coefficients are pooled across adjacent periods using time-dummy specifications. These models achieve relatively strong explanatory power in levels and produce smoother imputed price paths for non-overlapping products. The resulting attenuation of short-lived discounting effects generates materially higher measured price indices.

By contrast, price-change-based hedonic models yield cumulative price paths much closer to the traditional Jevons index, despite explicitly controlling for product characteristics. Because these models estimate price differences directly, they remain tightly anchored to observed price movements and limit the role of imputation across non-overlapping products. In this sense, quality adjustment alone does not mechanically reduce measured deflation. The econometric representation of price dynamics — levels versus changes, pooled versus locally estimated parameters — plays an equally decisive role.

Importantly, the results do not imply that hedonic methods systematically bias measured inflation upward or downward. In principle, bias could go either way: a new model may embody genuine quality improvements without increasing price, or conversely may raise price without proportional performance gains. The direction of divergence depends on how entry, exit, discounting, and imputation are handled in practice. This finding echoes the concerns emphasized by Melser and Syed (2016), who show that measured inflation in markets with frequent product replacement is highly sensitive to lifecycle pricing patterns and the treatment of product turnover rather than to any single index formula.

Taken together, the evidence suggests that disagreements between traditional and hedonic indices often reflect methodological choices rather than purely economic disagreement about quality change. Smartphones illustrate a broader lesson for high-technology price measurement: measured inflation is highly sensitive to how models treat product replacement, temporary promotions, and parameter stability across time.

More broadly, the analysis reinforces that no single index construction dominates across environments. Credible measurement in innovation-intensive markets requires triangulating across matched-model, level-based hedonic, and price-change-based hedonic approaches to assess robustness. As statistical agencies increasingly integrate alternative data sources and machine learning tools into official price statistics, transparency regarding model structure becomes essential. Methodological flexibility can materially influence measured economic outcomes, particularly in dynamic product markets.

Future research may extend this framework to other rapidly innovating consumer categories or evaluate the aggregate inflation implications of alternative index constructions. Where sufficiently rich expenditure data are available, superlative index formulas—such as Fisher or Törnqvist indices—may provide a preferable benchmark, as they offer a flexible approximation to the underlying cost-of-living index. More generally, this study underscores that advances in econometric techniques must be accompanied by careful economic interpretation when measuring price change in evolving product markets.

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A Additional Figures

Table 7: Level Regression Summary: 2020

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[0.0000, 0.2946]	0.0000	0.0752
Mobile Weight (grams) [SELECTED]	0.0054	[-0.0505, 0.0858]	0.9606	0.0348
RAM Memory (GB)	0.0000	[0.0000, 0.0387]	0.0000	0.0099
Front Camera (MP)	0.0000	[-0.0927, 0.1822]	0.0000	0.0701
Max Back Camera MP [SELECTED]	0.0205	[-0.3533, 0.1014]	0.9548	0.1160
Number of Cameras [SELECTED]	0.1895***	[0.0727, 0.2758]	0.0100	0.0518
Processor Level (encoded)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Battery Capacity (mAh) [SELECTED]	0.0783	[-0.0340, 0.3202]	0.7789	0.0903
Screen Size (inches)	0.0000	[0.0000, 0.0783]	0.0000	0.0200

Sample Size: 24 R^2 Score: 0.7211 Optimal Alpha: 0.020036 Features Selected: 4/9

[SELECTED] = Feature selected by Lasso Significance: *** $|\text{coef}| > 2 \cdot \text{SE}$, ** $|\text{coef}| > 1.96 \cdot \text{SE}$, * $|\text{coef}| > 1.645 \cdot \text{SE}$

Table 8: Level Regression Summary: 2022

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.5113***	[0.4265, 0.7902]	0.0100	0.0928
Mobile Weight (grams) [SELECTED]	0.0125	[-0.1900, 0.2007]	0.9680	0.0997
RAM Memory (GB) [SELECTED]	0.3538***	[0.2364, 0.6020]	0.0100	0.0933
Front Camera (MP) [SELECTED]	0.0307	[-0.0413, 0.1455]	0.8357	0.0477
Max Back Camera MP [SELECTED]	0.0206	[-0.1150, 0.1221]	0.9130	0.0605
Number of Cameras [SELECTED]	0.0071	[-0.0880, 0.1233]	0.9662	0.0539
Processor Level (encoded) [SELECTED]	0.1085*	[0.0000, 0.2378]	0.5438	0.0607
Battery Capacity (mAh) [SELECTED]	0.1207	[0.0000, 0.4233]	0.7149	0.1080
Screen Size (inches)	0.0000	[-0.2717, 0.0771]	0.0000	0.0890

Sample Size: 86 R^2 Score: 0.5427 Optimal Alpha: 0.012247 Features Selected: 8/9

[SELECTED] = Feature selected by Lasso Significance: *** $|coef| > 2 \cdot SE$, ** $|coef| > 1.96 \cdot SE$, * $|coef| > 1.645 \cdot SE$

Table 9: Level Regression Summary: 2023

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.4879***	[0.4193, 0.6396]	0.0100	0.0562
Mobile Weight (grams) [SELECTED]	0.0926	[-0.0618, 0.2013]	0.6479	0.0671
RAM Memory (GB) [SELECTED]	0.3859***	[0.2645, 0.5370]	0.0100	0.0695
Front Camera (MP) [SELECTED]	0.1275***	[0.0191, 0.2415]	0.0500	0.0567
Max Back Camera MP [SELECTED]	0.0267	[-0.0853, 0.1443]	0.8838	0.0586
Number of Cameras [SELECTED]	-0.0718*	[-0.1446, 0.0000]	0.5034	0.0369
Processor Level (encoded) [SELECTED]	0.0225	[-0.0606, 0.1258]	0.8794	0.0476
Battery Capacity (mAh)	0.0000	[-0.1216, 0.1524]	0.0000	0.0699
Screen Size (inches) [SELECTED]	0.0342	[-0.0814, 0.2087]	0.8821	0.0740

Sample Size: 118 R^2 Score: 0.5688 Optimal Alpha: 0.005697 Features Selected: 8/9

[SELECTED] = Feature selected by Lasso Significance: *** $|coef| > 2 \cdot SE$, ** $|coef| > 1.96 \cdot SE$, * $|coef| > 1.645 \cdot SE$

Table 10: Level Regression Summary: 2025

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.4443***	[0.3483, 0.6390]	0.0100	0.0742
Mobile Weight (grams)	0.0000	[-0.1706, 0.1200]	0.0000	0.0741
RAM Memory (GB) [SELECTED]	0.3857***	[0.2674, 0.5454]	0.0100	0.0709
Front Camera (MP) [SELECTED]	0.1428***	[0.0427, 0.2728]	0.0500	0.0587
Max Back Camera MP [SELECTED]	0.0575	[-0.0271, 0.1817]	0.7248	0.0533
Number of Cameras [SELECTED]	-0.0644*	[-0.1450, 0.0054]	0.5720	0.0384
Processor Level (encoded) [SELECTED]	-0.0565	[-0.1534, 0.0119]	0.6581	0.0422
Battery Capacity (mAh) [SELECTED]	0.1074	[-0.0040, 0.2798]	0.6216	0.0724
Screen Size (inches) [SELECTED]	0.0382	[-0.1069, 0.2481]	0.8924	0.0905

Sample Size: 148 R^2 Score: 0.5319 Optimal Alpha: 0.004409 Features Selected: 8/9

[SELECTED] = Feature selected by Lasso Significance: *** $|coef| > 2.5 \cdot SE$, ** $|coef| > 1.96 \cdot SE$, * $|coef| > 1.645 \cdot SE$

Table 11: Price-Change Regression Summary: 2021 → 2022

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[-0.1140, 0.0675]	0.0000	0.0463
Mobile Weight (grams) [SELECTED]	0.0043	[0.0000, 0.1640]	0.9736	0.0418
RAM Memory (GB) [SELECTED]	-0.0023	[-0.0423, 0.0604]	0.9777	0.0262
Front Camera (MP) [SELECTED]	-0.0123	[-0.0554, 0.0099]	0.8123	0.0167
Max Back Camera MP [SELECTED]	-0.0166	[-0.0503, 0.0000]	0.6698	0.0128
Number of Cameras [SELECTED]	-0.0373	[-0.1048, 0.0000]	0.6440	0.0267
Processor Level (encoded)	0.0000	[-0.0385, 0.0404]	0.0000	0.0201
Battery Capacity (mAh) [SELECTED]	0.0905	[0.0000, 0.2265]	0.6003	0.0578
Screen Size (inches) [SELECTED]	-0.0427	[-0.1922, 0.0000]	0.7777	0.0490

Sample Size: 47 R^2 Score: 0.4415 Optimal Alpha: 0.003135 Features Selected: 7/9

Table 12: Price-Change Regression Summary: 2023 → 2024

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[-0.0631, 0.0000]	0.0000	0.0161
Mobile Weight (grams)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
RAM Memory (GB)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Front Camera (MP)	0.0000	[0.0000, 0.0899]	0.0000	0.0229
Max Back Camera MP	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Number of Cameras	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Processor Level (encoded)	0.0000	[-0.0547, 0.0000]	0.0000	0.0140
Battery Capacity (mAh)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Screen Size (inches)	0.0000	[0.0000, 0.0000]	0.0000	0.0000

Sample Size: 118 R^2 Score: 0.0000 Optimal Alpha: 0.059111 Features Selected: 0/9

Table 13: Price-Change Regression Summary: 2024 → 2025

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	-0.0364***	[-0.0702, -0.0247]	0.0100	0.0116
Mobile Weight (grams)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
RAM Memory (GB)	0.0000	[0.0000, 0.0234]	0.0000	0.0060
Front Camera (MP) [SELECTED]	0.0025	[0.0000, 0.0394]	0.9375	0.0100
Max Back Camera MP	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Number of Cameras	0.0000	[0.0000, 0.0228]	0.0000	0.0058
Processor Level (encoded) [SELECTED]	-0.0028	[-0.0442, 0.0000]	0.9377	0.0113
Battery Capacity (mAh)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Screen Size (inches)	0.0000	[0.0000, 0.0046]	0.0000	0.0012

Sample Size: 138 R^2 Score: 0.1970 Optimal Alpha: 0.014774 Features Selected: 3/9

Table 14: Level Regression Summary 2020 Q1

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.1858	[-0.0540, 0.6051]	0.7182	0.1682
Mobile Weight (grams)	0.0000	[-0.0788, 0.2412]	0.0000	0.0816
RAM Memory (GB) [SELECTED]	0.2119	[0.0000, 0.6855]	0.6909	0.1749
Front Camera (MP) [SELECTED]	0.1634***	[0.0000, 0.2988]	0.5000	0.0762
Max Back Camera MP [SELECTED]	-0.1525	[-0.4053, 0.0000]	0.6237	0.1034
Number of Cameras [SELECTED]	0.2139***	[0.0816, 0.4100]	0.0500	0.0838
Processor Level (encoded)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Battery Capacity (mAh)	0.0000	[-0.4630, 0.3023]	0.0000	0.1952
Screen Size (inches) [SELECTED]	0.0618	[-0.0996, 0.3075]	0.8482	0.1038

Sample Size: 21 R^2 Score: 0.9175 Optimal Alpha: 0.001397 Features Selected: 6/9

[SELECTED] = Feature selected by Lasso Significance: *** $|\text{coef}| > 2{\cdot}\text{SE}$, ** $|\text{coef}| > 1.96{\cdot}\text{SE}$, * $|\text{coef}| > 1.645{\cdot}\text{SE}$

Table 15: Level Regression Summary 2020 Q2

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Mobile Weight (grams) [SELECTED]	0.0230	[-0.0151, 0.1241]	0.8349	0.0355
RAM Memory (GB)	0.0000	[0.0000, 0.1687]	0.0000	0.0430
Front Camera (MP) [SELECTED]	0.0842*	[0.0000, 0.1730]	0.5131	0.0441
Max Back Camera MP [SELECTED]	-0.0968***	[-0.1703, 0.0000]	0.5000	0.0435
Number of Cameras [SELECTED]	0.1423***	[0.0747, 0.2201]	0.0100	0.0371
Processor Level (encoded)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Battery Capacity (mAh) [SELECTED]	0.0495	[-0.1685, 0.0945]	0.8119	0.0671
Screen Size (inches)	0.0000	[0.0000, 0.1438]	0.0000	0.0367

Sample Size: 22 R^2 Score: 0.8957 Optimal Alpha: 0.011650 Features Selected: 5/9

[SELECTED] = Feature selected by Lasso Significance: *** $|\text{coef}| > 2{\cdot}\text{SE}$, ** $|\text{coef}| > 1.96{\cdot}\text{SE}$, * $|\text{coef}| > 1.645{\cdot}\text{SE}$

Table 16: Level Regression Summary 2020 Q3

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[0.0000, 0.2184]	0.0000	0.0557
Mobile Weight (grams)	0.0000	[-0.0044, 0.0689]	0.0000	0.0187
RAM Memory (GB)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Front Camera (MP)	0.0000	[-0.1245, 0.1478]	0.0000	0.0695
Max Back Camera MP	0.0000	[-0.2913, 0.0768]	0.0000	0.0939
Number of Cameras [SELECTED]	0.1253***	[0.0388, 0.2225]	0.0100	0.0469
Processor Level (encoded)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Battery Capacity (mAh) [SELECTED]	0.0837	[0.0000, 0.2628]	0.6813	0.0670
Screen Size (inches)	0.0000	[0.0000, 0.0000]	0.0000	0.0000

Sample Size: 24 R^2 Score: 0.5908 Optimal Alpha: 0.025959 Features Selected: 2/9

[SELECTED] = Feature selected by Lasso Significance: *** $|coef| > 2 \cdot SE$, ** $|coef| > 1.96 \cdot SE$, * $|coef| > 1.645 \cdot SE$

Table 17: Level Regression Summary 2020 Q4

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[0.0000, 0.3504]	0.0000	0.0894
Mobile Weight (grams)	0.0000	[-0.0006, 0.0882]	0.0000	0.0226
RAM Memory (GB)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Front Camera (MP)	0.0000	[-0.1495, 0.1719]	0.0000	0.0820
Max Back Camera MP [SELECTED]	0.0313	[-0.3112, 0.1269]	0.9284	0.1118
Number of Cameras [SELECTED]	0.1612***	[0.0458, 0.2815]	0.0100	0.0601
Processor Level (encoded)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Battery Capacity (mAh) [SELECTED]	0.0683	[-0.0043, 0.3266]	0.7937	0.0844
Screen Size (inches)	0.0000	[0.0000, 0.0000]	0.0000	0.0000

Sample Size: 24 R^2 Score: 0.5934 Optimal Alpha: 0.026642 Features Selected: 3/9

[SELECTED] = Feature selected by Lasso Significance: *** $|coef| > 2 \cdot SE$, ** $|coef| > 1.96 \cdot SE$, * $|coef| > 1.645 \cdot SE$

Table 18: Level Regression Summary 2021 Q1

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[0.0000, 0.1408]	0.0000	0.0359
Mobile Weight (grams)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
RAM Memory (GB)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Front Camera (MP)	0.0000	[0.0000, 0.2635]	0.0000	0.0672
Max Back Camera MP	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Number of Cameras	0.0000	[0.0000, 0.4133]	0.0000	0.1054
Processor Level (encoded)	0.0000	[-0.1561, 0.0000]	0.0000	0.0398
Battery Capacity (mAh)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Screen Size (inches)	0.0000	[0.0000, 0.0000]	0.0000	0.0000

Sample Size: 37 R^2 Score: 0.0000 Optimal Alpha: 0.230929 Features Selected: 0/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 19: Level Regression Summary 2021 Q2

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.9314***	[0.3777, 1.7800]	0.0100	0.3577
Mobile Weight (grams) [SELECTED]	-0.5065***	[-0.8928, 0.0310]	0.5000	0.2357
RAM Memory (GB) [SELECTED]	0.3867***	[0.1573, 0.8496]	0.0500	0.1766
Front Camera (MP) [SELECTED]	0.0849	[-0.0377, 0.3317]	0.7701	0.0942
Max Back Camera MP [SELECTED]	0.1812	[-0.2900, 0.3968]	0.7361	0.1752
Number of Cameras [SELECTED]	0.2488***	[0.0575, 0.4115]	0.0100	0.0903
Processor Level (encoded) [SELECTED]	-0.0241	[-0.1173, 0.0912]	0.8843	0.0532
Battery Capacity (mAh) [SELECTED]	0.2033	[-0.4248, 0.6530]	0.8114	0.2750
Screen Size (inches) [SELECTED]	0.3271***	[0.0897, 0.7092]	0.0500	0.1580

Sample Size: 40 R^2 Score: 0.8173 Optimal Alpha: 0.000226 Features Selected: 9/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2.5·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 20: Level Regression Summary 2021 Q3

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.8732***	[0.5186, 1.6799]	0.0100	0.2963
Mobile Weight (grams) [SELECTED]	-0.3727*	[-0.8686, 0.0000]	0.5710	0.2216
RAM Memory (GB) [SELECTED]	0.4020***	[0.2218, 0.7343]	0.0100	0.1307
Front Camera (MP) [SELECTED]	0.0751	[-0.0775, 0.2505]	0.7712	0.0837
Max Back Camera MP [SELECTED]	0.1564	[-0.2040, 0.3192]	0.7009	0.1334
Number of Cameras [SELECTED]	0.2240***	[0.0833, 0.3932]	0.0100	0.0790
Processor Level (encoded) [SELECTED]	-0.0099	[-0.1478, 0.1693]	0.9688	0.0809
Battery Capacity (mAh) [SELECTED]	0.1078	[-0.2912, 0.8108]	0.9022	0.2811
Screen Size (inches) [SELECTED]	0.2926***	[0.1139, 0.6478]	0.0500	0.1362

Sample Size: 43 R^2 Score: 0.8548 Optimal Alpha: 0.004321 Features Selected: 9/9

[SELECTED] = Feature selected by Lasso Significance: *** $|\text{coef}| > 2\cdot\text{SE}$, ** $|\text{coef}| > 1.96\cdot\text{SE}$, * $|\text{coef}| > 1.645\cdot\text{SE}$

Table 21: Level Regression Summary 2021 Q4

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	1.0618***	[0.6814, 1.6289]	0.0100	0.2417
Mobile Weight (grams) [SELECTED]	-0.4530***	[-0.8306, -0.0860]	0.0500	0.1900
RAM Memory (GB) [SELECTED]	0.4696***	[0.2887, 0.7099]	0.0100	0.1074
Front Camera (MP) [SELECTED]	0.0448	[-0.0809, 0.2002]	0.8406	0.0717
Max Back Camera MP [SELECTED]	0.1650	[-0.3019, 0.3271]	0.7377	0.1605
Number of Cameras [SELECTED]	0.1941***	[0.0682, 0.3438]	0.0100	0.0703
Processor Level (encoded)	0.0000	[-0.1422, 0.1916]	0.0000	0.0852
Battery Capacity (mAh) [SELECTED]	0.3688*	[0.0000, 0.7646]	0.5177	0.1950
Screen Size (inches) [SELECTED]	0.1973	[0.0000, 0.4770]	0.5865	0.1217

Sample Size: 47 R^2 Score: 0.8597 Optimal Alpha: 0.002201 Features Selected: 8/9

[SELECTED] = Feature selected by Lasso Significance: *** $|\text{coef}| > 2\cdot\text{SE}$, ** $|\text{coef}| > 1.96\cdot\text{SE}$, * $|\text{coef}| > 1.645\cdot\text{SE}$

Table 22: Level Regression Summary 2022 Q1

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	1.1732***	[0.8536, 1.6599]	0.0100	0.2057
Mobile Weight (grams) [SELECTED]	-0.4700***	[-0.6761, -0.2586]	0.0100	0.1065
RAM Memory (GB) [SELECTED]	0.5895***	[0.3826, 0.8399]	0.0100	0.1167
Front Camera (MP) [SELECTED]	0.0419	[-0.0888, 0.2134]	0.8612	0.0771
Max Back Camera MP [SELECTED]	0.1962	[-0.1335, 0.3576]	0.6005	0.1253
Number of Cameras [SELECTED]	0.2209***	[0.0680, 0.3766]	0.0100	0.0787
Processor Level (encoded) [SELECTED]	0.0049	[-0.1094, 0.1549]	0.9813	0.0674
Battery Capacity (mAh) [SELECTED]	0.5989***	[0.2353, 0.9099]	0.0100	0.1721
Screen Size (inches) [SELECTED]	-0.0227	[-0.2749, 0.2366]	0.9556	0.1305

Sample Size: 61 R^2 Score: 0.7111 Optimal Alpha: 0.000264 Features Selected: 9/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2.5·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 23: Level Regression Summary 2022 Q2

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[0.0000, 0.2121]	0.0000	0.0541
Mobile Weight (grams)	0.0000	[0.0000, 0.1284]	0.0000	0.0327
RAM Memory (GB)	0.0000	[0.0000, 0.1342]	0.0000	0.0342
Front Camera (MP)	0.0000	[0.0000, 0.1280]	0.0000	0.0327
Max Back Camera MP	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Number of Cameras [SELECTED]	0.0522	[0.0000, 0.3267]	0.8401	0.0833
Processor Level (encoded)	0.0000	[0.0000, 0.2654]	0.0000	0.0677
Battery Capacity (mAh)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Screen Size (inches)	0.0000	[0.0000, 0.0000]	0.0000	0.0000

Sample Size: 67 R^2 Score: 0.0752 Optimal Alpha: 0.162286 Features Selected: 1/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 24: Level Regression Summary 2022 Q3

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.4459***	[0.3443, 1.2004]	0.0500	0.2184
Mobile Weight (grams)	0.0000	[-0.4726, 0.2027]	0.0000	0.1723
RAM Memory (GB) [SELECTED]	0.2860***	[0.1944, 0.6262]	0.0100	0.1102
Front Camera (MP) [SELECTED]	0.0386	[-0.0964, 0.1710]	0.8555	0.0682
Max Back Camera MP [SELECTED]	0.0725	[0.0000, 0.2227]	0.6743	0.0568
Number of Cameras [SELECTED]	0.1223*	[-0.0149, 0.2605]	0.5557	0.0703
Processor Level (encoded) [SELECTED]	0.0992*	[0.0000, 0.2323]	0.5731	0.0593
Battery Capacity (mAh) [SELECTED]	0.1006	[0.0000, 0.8027]	0.8746	0.2048
Screen Size (inches) [SELECTED]	-0.1023	[-0.5512, 0.0000]	0.8143	0.1406

Sample Size: 71 R^2 Score: 0.5337 Optimal Alpha: 0.015476 Features Selected: 8/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 25: Level Regression Summary 2022 Q4

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.5644***	[0.4635, 0.8493]	0.0100	0.0984
Mobile Weight (grams) [SELECTED]	0.0028	[-0.2082, 0.1993]	0.9931	0.1039
RAM Memory (GB) [SELECTED]	0.3904***	[0.2702, 0.6279]	0.0100	0.0912
Front Camera (MP) [SELECTED]	0.0627	[-0.0035, 0.2016]	0.6941	0.0523
Max Back Camera MP	0.0000	[-0.1652, 0.0978]	0.0000	0.0671
Number of Cameras [SELECTED]	-0.0191	[-0.1636, 0.0520]	0.9112	0.0550
Processor Level (encoded) [SELECTED]	0.1183**	[0.0165, 0.2526]	0.0500	0.0602
Battery Capacity (mAh) [SELECTED]	0.1502	[0.0000, 0.4586]	0.6725	0.1170
Screen Size (inches)	0.0000	[-0.2507, 0.1141]	0.0000	0.0930

Sample Size: 86 R^2 Score: 0.5740 Optimal Alpha: 0.010247 Features Selected: 7/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 26: Level Regression Summary 2023 Q1

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.4903***	[0.4430, 0.7484]	0.0100	0.0779
Mobile Weight (grams) [SELECTED]	0.1470**	[-0.0335, 0.2581]	0.5000	0.0744
RAM Memory (GB) [SELECTED]	0.3931***	[0.2671, 0.5845]	0.0100	0.0810
Front Camera (MP) [SELECTED]	0.1344*	[0.0000, 0.2829]	0.5248	0.0722
Max Back Camera MP [SELECTED]	0.0042	[-0.1384, 0.1535]	0.9855	0.0745
Number of Cameras [SELECTED]	-0.0741	[-0.1906, 0.0191]	0.6467	0.0535
Processor Level (encoded) [SELECTED]	0.0711	[0.0000, 0.1859]	0.6176	0.0474
Battery Capacity (mAh) [SELECTED]	0.0195	[-0.1016, 0.2793]	0.9489	0.0972
Screen Size (inches) [SELECTED]	0.0032	[-0.1834, 0.2283]	0.9922	0.1050

Sample Size: 92 R^2 Score: 0.5713   Optimal Alpha: 0.009505   Features Selected: 9/9

[SELECTED] = Feature selected by Lasso   Significance: *** |coef| > 2.5·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 27: Level Regression Summary 2023 Q2

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.2896***	[0.3406, 0.6184]	0.0100	0.0709
Mobile Weight (grams) [SELECTED]	0.0956	[-0.0804, 0.2019]	0.6612	0.0720
RAM Memory (GB) [SELECTED]	0.2653***	[0.2053, 0.5115]	0.0100	0.0781
Front Camera (MP) [SELECTED]	0.0609	[0.0000, 0.2528]	0.7591	0.0645
Max Back Camera MP	0.0000	[0.0000, 0.1687]	0.0000	0.0430
Number of Cameras	0.0000	[-0.1512, 0.0229]	0.0000	0.0444
Processor Level (encoded)	0.0000	[-0.0582, 0.1237]	0.0000	0.0464
Battery Capacity (mAh)	0.0000	[0.0000, 0.2198]	0.0000	0.0561
Screen Size (inches)	0.0000	[0.0000, 0.1832]	0.0000	0.0467

Sample Size: 99 R^2 Score: 0.4452   Optimal Alpha: 0.040742   Features Selected: 4/9

[SELECTED] = Feature selected by Lasso   Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 28: Level Regression Summary 2023 Q3

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.3556***	[0.3536, 0.6214]	0.0100	0.0683
Mobile Weight (grams) [SELECTED]	0.0914	[-0.0440, 0.2066]	0.6355	0.0640
RAM Memory (GB) [SELECTED]	0.3299***	[0.2519, 0.5307]	0.0100	0.0711
Front Camera (MP) [SELECTED]	0.0774	[0.0000, 0.2346]	0.6702	0.0598
Max Back Camera MP	0.0000	[0.0000, 0.1710]	0.0000	0.0436
Number of Cameras	0.0000	[-0.1242, 0.0463]	0.0000	0.0435
Processor Level (encoded)	0.0000	[-0.0863, 0.0919]	0.0000	0.0455
Battery Capacity (mAh)	0.0000	[-0.1493, 0.1208]	0.0000	0.0689
Screen Size (inches)	0.0000	[0.0000, 0.2205]	0.0000	0.0563

Sample Size: 102 R^2 Score: 0.5155 Optimal Alpha: 0.026714 Features Selected: 4/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 29: Level Regression Summary 2023 Q4

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.4723***	[0.4155, 0.6463]	0.0100	0.0589
Mobile Weight (grams) [SELECTED]	0.0701	[-0.0174, 0.1769]	0.6389	0.0496
RAM Memory (GB) [SELECTED]	0.3742***	[0.2608, 0.5231]	0.0100	0.0669
Front Camera (MP) [SELECTED]	0.1403***	[0.0282, 0.2611]	0.0500	0.0594
Max Back Camera MP [SELECTED]	0.0510	[-0.0649, 0.1730]	0.7858	0.0607
Number of Cameras [SELECTED]	-0.0680*	[-0.1566, 0.0000]	0.5656	0.0400
Processor Level (encoded) [SELECTED]	0.0047	[-0.0902, 0.1203]	0.9777	0.0537
Battery Capacity (mAh)	0.0000	[-0.1216, 0.1741]	0.0000	0.0754
Screen Size (inches) [SELECTED]	0.0201	[-0.1156, 0.1825]	0.9326	0.0760

Sample Size: 117 R^2 Score: 0.5587 Optimal Alpha: 0.008342 Features Selected: 8/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2.5·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 30: Level Regression Summary 2024 Q1

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.4310***	[0.3534, 0.5983]	0.0100	0.0625
Mobile Weight (grams) [SELECTED]	0.0631	[-0.0933, 0.1904]	0.7777	0.0724
RAM Memory (GB) [SELECTED]	0.3950***	[0.2726, 0.5556]	0.0100	0.0722
Front Camera (MP) [SELECTED]	0.1063*	[0.0000, 0.2410]	0.5590	0.0615
Max Back Camera MP [SELECTED]	0.0332	[-0.0866, 0.1466]	0.8574	0.0595
Number of Cameras [SELECTED]	-0.0832*	[-0.1783, 0.0000]	0.5333	0.0455
Processor Level (encoded) [SELECTED]	-0.0144	[-0.1103, 0.0784]	0.9239	0.0481
Battery Capacity (mAh) [SELECTED]	0.0334	[-0.0908, 0.2325]	0.8967	0.0825
Screen Size (inches)	0.0000	[-0.1498, 0.1513]	0.0000	0.0768

Sample Size: 121 R^2 Score: 0.5062 Optimal Alpha: 0.007584 Features Selected: 8/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2.5·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 31: Level Regression Summary 2024 Q2

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.4563***	[0.3628, 0.6407]	0.0100	0.0709
Mobile Weight (grams) [SELECTED]	0.0460	[-0.1423, 0.1599]	0.8477	0.0771
RAM Memory (GB) [SELECTED]	0.3699***	[0.2192, 0.5280]	0.0100	0.0788
Front Camera (MP) [SELECTED]	0.1594***	[0.0494, 0.2974]	0.0500	0.0633
Max Back Camera MP [SELECTED]	0.0702	[-0.0438, 0.1891]	0.6988	0.0594
Number of Cameras [SELECTED]	-0.0638	[-0.1422, 0.0149]	0.5940	0.0401
Processor Level (encoded) [SELECTED]	-0.0206	[-0.1309, 0.0648]	0.8947	0.0499
Battery Capacity (mAh) [SELECTED]	0.0512	[-0.0759, 0.2475]	0.8418	0.0825
Screen Size (inches) [SELECTED]	0.0225	[-0.1250, 0.1935]	0.9295	0.0813

Sample Size: 129 R^2 Score: 0.4984 Optimal Alpha: 0.004065 Features Selected: 9/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2.5·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 32: Level Regression Summary 2024 Q3

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.4255***	[0.3579, 0.5998]	0.0100	0.0617
Mobile Weight (grams) [SELECTED]	0.0550	[-0.0919, 0.1645]	0.7854	0.0654
RAM Memory (GB) [SELECTED]	0.3574***	[0.2248, 0.5266]	0.0100	0.0770
Front Camera (MP) [SELECTED]	0.1460***	[0.0319, 0.2875]	0.0500	0.0652
Max Back Camera MP [SELECTED]	0.0566	[0.0000, 0.1657]	0.6583	0.0423
Number of Cameras [SELECTED]	-0.0373	[-0.1261, 0.0185]	0.7421	0.0369
Processor Level (encoded) [SELECTED]	-0.0057	[-0.1218, 0.0713]	0.9704	0.0493
Battery Capacity (mAh) [SELECTED]	0.0236	[-0.1127, 0.2165]	0.9284	0.0840
Screen Size (inches) [SELECTED]	0.0139	[-0.1083, 0.1616]	0.9485	0.0689

Sample Size: 128 R^2 Score: 0.4950 Optimal Alpha: 0.009347 Features Selected: 9/9

[SELECTED] = Feature selected by Lasso Significance: *** $|coef| > 2 \cdot SE$, ** $|coef| > 1.96 \cdot SE$, * $|coef| > 1.645 \cdot SE$

Table 33: Level Regression Summary 2024 Q4

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.4827***	[0.3731, 0.6366]	0.0100	0.0672
Mobile Weight (grams) [SELECTED]	0.0444	[-0.1404, 0.1693]	0.8565	0.0790
RAM Memory (GB) [SELECTED]	0.3836***	[0.2451, 0.5358]	0.0100	0.0742
Front Camera (MP) [SELECTED]	0.1423***	[0.0225, 0.2735]	0.0500	0.0640
Max Back Camera MP [SELECTED]	0.0483	[-0.0546, 0.1703]	0.7855	0.0574
Number of Cameras [SELECTED]	-0.0932***	[-0.1692, -0.0029]	0.0500	0.0424
Processor Level (encoded) [SELECTED]	-0.0518	[-0.1564, 0.0414]	0.7381	0.0505
Battery Capacity (mAh) [SELECTED]	0.0871	[-0.0441, 0.2732]	0.7254	0.0810
Screen Size (inches)	0.0000	[-0.1475, 0.1696]	0.0000	0.0809

Sample Size: 140 R^2 Score: 0.5223 Optimal Alpha: 0.001761 Features Selected: 8/9

[SELECTED] = Feature selected by Lasso Significance: *** $|coef| > 2 \cdot SE$, ** $|coef| > 1.96 \cdot SE$, * $|coef| > 1.645 \cdot SE$

Table 34: Level Regression Summary 2025 Q1

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.4525***	[0.3436, 0.6335]	0.0100	0.0739
Mobile Weight (grams)	0.0000	[-0.1747, 0.1261]	0.0000	0.0767
RAM Memory (GB) [SELECTED]	0.3861***	[0.2610, 0.5243]	0.0100	0.0672
Front Camera (MP) [SELECTED]	0.1336***	[0.0170, 0.2673]	0.0500	0.0638
Max Back Camera MP [SELECTED]	0.0624	[-0.0290, 0.1876]	0.7120	0.0553
Number of Cameras [SELECTED]	-0.0618	[-0.1455, 0.0276]	0.6429	0.0441
Processor Level (encoded) [SELECTED]	-0.0460	[-0.1496, 0.0256]	0.7375	0.0447
Battery Capacity (mAh) [SELECTED]	0.1115	[-0.0416, 0.2752]	0.6480	0.0808
Screen Size (inches) [SELECTED]	0.0310	[-0.1171, 0.2305]	0.9108	0.0887

Sample Size: 146 R^2 Score: 0.5258 Optimal Alpha: 0.003013 Features Selected: 8/9
[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 35: Level Regression Summary 2025 Q2

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.5250***	[0.3985, 0.6637]	0.0100	0.0676
Mobile Weight (grams) [SELECTED]	-0.0949	[-0.2850, 0.0171]	0.6858	0.0771
RAM Memory (GB) [SELECTED]	0.3934***	[0.1999, 0.5549]	0.0100	0.0906
Front Camera (MP) [SELECTED]	0.1324*	[-0.0235, 0.2596]	0.5324	0.0722
Max Back Camera MP [SELECTED]	0.0710	[-0.0607, 0.2081]	0.7358	0.0686
Number of Cameras [SELECTED]	-0.0554	[-0.1230, 0.0363]	0.6523	0.0406
Processor Level (encoded) [SELECTED]	-0.0698	[-0.1433, 0.0280]	0.5927	0.0437
Battery Capacity (mAh) [SELECTED]	0.1286	[-0.0242, 0.3382]	0.6451	0.0924
Screen Size (inches) [SELECTED]	0.1704***	[0.0111, 0.3344]	0.0500	0.0825

Sample Size: 133 R^2 Score: 0.5776 Optimal Alpha: 0.000252 Features Selected: 9/9
[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2.5·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 36: Price-Change Regression Summary 2020 Q1 → 2020 Q2

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[-0.0253, 0.0000]	0.0000	0.0064
Mobile Weight (grams) [SELECTED]	-0.0004	[-0.0407, 0.0000]	0.9892	0.0104
RAM Memory (GB)	0.0000	[-0.0252, 0.0000]	0.0000	0.0064
Front Camera (MP) [SELECTED]	-0.0434*	[-0.0920, 0.0000]	0.5285	0.0235
Max Back Camera MP	0.0000	[-0.0219, 0.0000]	0.0000	0.0056
Number of Cameras [SELECTED]	-0.0436*	[-0.0998, 0.0000]	0.5629	0.0255
Processor Level (encoded)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Battery Capacity (mAh)	0.0000	[-0.0277, 0.0000]	0.0000	0.0071
Screen Size (inches)	0.0000	[0.0000, 0.0000]	0.0000	0.0000

Sample Size: 21 R^2 Score: 0.6820 Optimal Alpha: 0.015134 Features Selected: 3/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 37: Price-Change Regression Summary 2020 Q2 → 2020 Q3

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[-0.0472, 0.0000]	0.0000	0.0120
Mobile Weight (grams)	0.0000	[-0.0358, 0.0000]	0.0000	0.0091
RAM Memory (GB)	0.0000	[0.0000, 0.0450]	0.0000	0.0115
Front Camera (MP) [SELECTED]	-0.0085	[-0.0491, 0.0074]	0.8494	0.0144
Max Back Camera MP	0.0000	[-0.0497, 0.0000]	0.0000	0.0127
Number of Cameras [SELECTED]	-0.0308**	[-0.0613, 0.0000]	0.5000	0.0156
Processor Level (encoded)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Battery Capacity (mAh)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Screen Size (inches)	0.0000	[0.0000, 0.0323]	0.0000	0.0082

Sample Size: 22 R^2 Score: 0.4243 Optimal Alpha: 0.017068 Features Selected: 2/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 38: Price-Change Regression Summary 2020 Q3 → 2020 Q4

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[0.0000, 0.0442]	0.0000	0.0113
Mobile Weight (grams)	0.0000	[0.0000, 0.0203]	0.0000	0.0052
RAM Memory (GB)	0.0000	[-0.0413, 0.0000]	0.0000	0.0105
Front Camera (MP)	0.0000	[0.0000, 0.0195]	0.0000	0.0050
Max Back Camera MP [SELECTED]	0.0007	[0.0000, 0.0546]	0.9867	0.0139
Number of Cameras [SELECTED]	0.0071	[0.0000, 0.0445]	0.8396	0.0113
Processor Level (encoded)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Battery Capacity (mAh)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Screen Size (inches)	0.0000	[0.0000, 0.0000]	0.0000	0.0000

Sample Size: 24 R^2 Score: 0.1129 Optimal Alpha: 0.021688 Features Selected: 2/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 39: Price-Change Regression Summary 2020 Q4 → 2021 Q1

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	-0.0487	[-0.3315, 0.0000]	0.8531	0.0846
Mobile Weight (grams) [SELECTED]	0.0341	[-0.0070, 0.1061]	0.6985	0.0289
RAM Memory (GB) [SELECTED]	-0.0138	[-0.2948, 0.0000]	0.9533	0.0752
Front Camera (MP) [SELECTED]	0.0627***	[0.0304, 0.1053]	0.0100	0.0191
Max Back Camera MP [SELECTED]	-0.0119	[-0.0874, 0.0617]	0.9204	0.0380
Number of Cameras [SELECTED]	-0.0245	[-0.0586, 0.0019]	0.5954	0.0154
Processor Level (encoded)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Battery Capacity (mAh) [SELECTED]	-0.0614	[-0.1748, 0.0000]	0.6486	0.0446
Screen Size (inches)	0.0000	[-0.0382, 0.0689]	0.0000	0.0273

Sample Size: 24 R^2 Score: 0.7149 Optimal Alpha: 0.001345 Features Selected: 7/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 40: Price-Change Regression Summary 2020 Q4 → 2021 Q1

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	-0.0487	[-0.3315, 0.0000]	0.8531	0.0846
Mobile Weight (grams) [SELECTED]	0.0341	[-0.0070, 0.1061]	0.6985	0.0289
RAM Memory (GB) [SELECTED]	-0.0138	[-0.2948, 0.0000]	0.9533	0.0752
Front Camera (MP) [SELECTED]	0.0627***	[0.0304, 0.1053]	0.0100	0.0191
Max Back Camera MP [SELECTED]	-0.0119	[-0.0874, 0.0617]	0.9204	0.0380
Number of Cameras [SELECTED]	-0.0245	[-0.0586, 0.0019]	0.5954	0.0154
Processor Level (encoded)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Battery Capacity (mAh) [SELECTED]	-0.0614	[-0.1748, 0.0000]	0.6486	0.0446
Screen Size (inches)	0.0000	[-0.0382, 0.0689]	0.0000	0.0273

Sample Size: 24 R^2 Score: 0.7149 Optimal Alpha: 0.001345 Features Selected: 7/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 41: Price-Change Regression Summary 2021 Q2 → 2021 Q3

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[0.0000, 0.0109]	0.0000	0.0028
Mobile Weight (grams)	0.0000	[-0.0169, 0.0223]	0.0000	0.0100
RAM Memory (GB)	0.0000	[-0.0264, 0.0360]	0.0000	0.0159
Front Camera (MP) [SELECTED]	-0.0297***	[-0.0608, -0.0203]	0.0100	0.0103
Max Back Camera MP	0.0000	[-0.0448, 0.0000]	0.0000	0.0114
Number of Cameras [SELECTED]	0.0039	[0.0000, 0.0244]	0.8394	0.0062
Processor Level (encoded)	0.0000	[-0.0035, 0.0000]	0.0000	0.0009
Battery Capacity (mAh)	0.0000	[0.0000, 0.0595]	0.0000	0.0152
Screen Size (inches) [SELECTED]	0.0088	[-0.0169, 0.0335]	0.8261	0.0129

Sample Size: 40 R^2 Score: 0.4167 Optimal Alpha: 0.008311 Features Selected: 3/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 42: Price-Change Regression Summary 2021 Q3 → 2021 Q4

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[-0.0229, 0.0086]	0.0000	0.0080
Mobile Weight (grams)	0.0000	[-0.0099, 0.0306]	0.0000	0.0103
RAM Memory (GB)	0.0000	[-0.0557, 0.0000]	0.0000	0.0142
Front Camera (MP)	0.0000	[-0.0246, 0.0000]	0.0000	0.0063
Max Back Camera MP	0.0000	[0.0000, 0.0280]	0.0000	0.0072
Number of Cameras [SELECTED]	-0.0229	[-0.0664, 0.0000]	0.6561	0.0169
Processor Level (encoded) [SELECTED]	-0.0000	[-0.0176, 0.0000]	0.9974	0.0045
Battery Capacity (mAh) [SELECTED]	0.0118	[0.0000, 0.0735]	0.8399	0.0188
Screen Size (inches)	0.0000	[0.0000, 0.0145]	0.0000	0.0037

Sample Size: 43 R^2 Score: 0.2560 Optimal Alpha: 0.008366 Features Selected: 3/9
[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2.5·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 43: Price-Change Regression Summary 2021 Q4 → 2022 Q1

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	-0.0393***	[-0.0714, 0.0020]	0.5000	0.0187
Mobile Weight (grams)	0.0000	[-0.0252, 0.0000]	0.0000	0.0064
RAM Memory (GB) [SELECTED]	0.0007	[0.0000, 0.0693]	0.9898	0.0177
Front Camera (MP)	0.0000	[0.0000, 0.0228]	0.0000	0.0058
Max Back Camera MP	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Number of Cameras	0.0000	[-0.0240, 0.0000]	0.0000	0.0061
Processor Level (encoded)	0.0000	[-0.0310, 0.0000]	0.0000	0.0079
Battery Capacity (mAh)	0.0000	[0.0000, 0.0400]	0.0000	0.0102
Screen Size (inches)	0.0000	[0.0000, 0.0000]	0.0000	0.0000

Sample Size: 47 R^2 Score: 0.4053 Optimal Alpha: 0.019396 Features Selected: 2/9
[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 44: Price-Change Regression Summary 2022 Q1 → 2022 Q2

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[0.0000, 0.0216]	0.0000	0.0055
Mobile Weight (grams)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
RAM Memory (GB)	-0.0000	[-0.0413, 0.0000]	0.0000	0.0105
Front Camera (MP)	0.0000	[-0.0282, 0.0366]	0.0000	0.0165
Max Back Camera MP	0.0000	[-0.0182, 0.0000]	0.0000	0.0047
Number of Cameras	0.0000	[0.0000, 0.0084]	0.0000	0.0021
Processor Level (encoded)	0.0000	[0.0000, 0.0308]	0.0000	0.0079
Battery Capacity (mAh)	0.0000	[-0.0124, 0.0000]	0.0000	0.0032
Screen Size (inches)	0.0000	[-0.0077, 0.0000]	0.0000	0.0020

Sample Size: 61 R^2 Score: 0.0000 Optimal Alpha: 0.018138 Features Selected: 0/9
[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 45: Price-Change Regression Summary 2022 Q2 → 2022 Q3

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	0.0226	[-0.0111, 0.0557]	0.6620	0.0171
Mobile Weight (grams)	0.0000	[-0.0347, 0.0000]	0.0000	0.0088
RAM Memory (GB) [SELECTED]	-0.0165	[-0.0549, 0.0000]	0.6999	0.0140
Front Camera (MP)	0.0000	[-0.0243, 0.0296]	0.0000	0.0137
Max Back Camera MP [SELECTED]	-0.0136	[-0.0383, 0.0000]	0.6451	0.0098
Number of Cameras	0.0000	[-0.0124, 0.0292]	0.0000	0.0106
Processor Level (encoded)	0.0000	[-0.0199, 0.0000]	0.0000	0.0051
Battery Capacity (mAh) [SELECTED]	-0.0103	[-0.0464, 0.0085]	0.8115	0.0140
Screen Size (inches)	0.0000	[-0.0253, 0.0000]	0.0000	0.0064

Sample Size: 67 R^2 Score: 0.4830 Optimal Alpha: 0.006906 Features Selected: 4/9
[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 46: Price-Change Regression Summary 2022 Q3 → 2022 Q4

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[-0.0424, 0.0000]	0.0000	0.0108
Mobile Weight (grams)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
RAM Memory (GB)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Front Camera (MP)	0.0000	[0.0000, 0.0685]	0.0000	0.0175
Max Back Camera MP	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Number of Cameras	0.0000	[-0.0532, 0.0000]	0.0000	0.0136
Processor Level (encoded)	0.0000	[-0.0410, 0.0000]	0.0000	0.0105
Battery Capacity (mAh)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Screen Size (inches)	0.0000	[0.0000, 0.0000]	0.0000	0.0000

Sample Size: 71 R^2 Score: 0.0000 Optimal Alpha: 0.047572 Features Selected: 0/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2.5·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 47: Price-Change Regression Summary 2022 Q4 → 2023 Q1

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[-0.0405, 0.0000]	0.0000	0.0103
Mobile Weight (grams)	0.0000	[0.0000, 0.0479]	0.0000	0.0122
RAM Memory (GB) [SELECTED]	0.0060	[0.0000, 0.0460]	0.8705	0.0117
Front Camera (MP)	0.0000	[-0.0441, 0.0000]	0.0000	0.0112
Max Back Camera MP	0.0000	[0.0000, 0.0140]	0.0000	0.0036
Number of Cameras	0.0000	[0.0000, 0.0286]	0.0000	0.0073
Processor Level (encoded)	0.0000	[0.0000, 0.0164]	0.0000	0.0042
Battery Capacity (mAh)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Screen Size (inches) [SELECTED]	0.0284	[-0.0199, 0.0551]	0.6215	0.0191

Sample Size: 85 R^2 Score: 0.3068 Optimal Alpha: 0.014186 Features Selected: 2/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 48: Price-Change Regression Summary 2023 Q1 → 2023 Q2

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[0.0000, 0.0090]	0.0000	0.0023
Mobile Weight (grams)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
RAM Memory (GB) [SELECTED]	-0.0069	[-0.0439, 0.0000]	0.8437	0.0112
Front Camera (MP)	0.0000	[-0.0075, 0.0445]	0.0000	0.0133
Max Back Camera MP [SELECTED]	-0.0157	[-0.0399, 0.0000]	0.6056	0.0102
Number of Cameras	0.0000	[0.0000, 0.0263]	0.0000	0.0067
Processor Level (encoded)	0.0000	[-0.0227, 0.0000]	0.0000	0.0058
Battery Capacity (mAh) [SELECTED]	-0.0038	[-0.0381, 0.0000]	0.9012	0.0097
Screen Size (inches)	0.0000	[-0.0236, 0.0000]	0.0000	0.0060

Sample Size: 90 R^2 Score: 0.1930 Optimal Alpha: 0.012409 Features Selected: 3/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 49: Price-Change Regression Summary 2023 Q2 → 2023 Q3

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Mobile Weight (grams)	0.0000	[-0.0259, 0.0000]	0.0000	0.0066
RAM Memory (GB)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Front Camera (MP)	0.0000	[-0.0192, 0.0000]	0.0000	0.0049
Max Back Camera MP	0.0000	[-0.0291, 0.0000]	0.0000	0.0074
Number of Cameras	0.0000	[0.0000, 0.0348]	0.0000	0.0089
Processor Level (encoded) [SELECTED]	-0.0021	[-0.0567, 0.0000]	0.9632	0.0145
Battery Capacity (mAh)	0.0000	[-0.0263, 0.0000]	0.0000	0.0067
Screen Size (inches)	0.0000	[-0.0209, 0.0000]	0.0000	0.0053

Sample Size: 99 R^2 Score: 0.0138 Optimal Alpha: 0.028869 Features Selected: 1/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 50: Price-Change Regression Summary 2023 Q3 → 2023 Q4

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	-0.0081	[-0.0616, 0.0000]	0.8688	0.0157
Mobile Weight (grams) [SELECTED]	-0.0193	[-0.0524, 0.0148]	0.7133	0.0171
RAM Memory (GB)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Front Camera (MP) [SELECTED]	0.0106	[0.0000, 0.0445]	0.7618	0.0113
Max Back Camera MP [SELECTED]	0.0108	[0.0000, 0.0568]	0.8104	0.0145
Number of Cameras	0.0000	[-0.0118, 0.0000]	0.0000	0.0030
Processor Level (encoded)	0.0000	[-0.0290, 0.0163]	0.0000	0.0116
Battery Capacity (mAh)	0.0000	[0.0000, 0.0371]	0.0000	0.0095
Screen Size (inches)	0.0000	[-0.0758, 0.0000]	0.0000	0.0193

Sample Size: 102 R^2 Score: 0.2180   Optimal Alpha: 0.014017   Features Selected: 4/9

[SELECTED] = Feature selected by Lasso   Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 51: Price-Change Regression Summary 2023 Q4 → 2024 Q1

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	-0.0146	[-0.0448, 0.0000]	0.6734	0.0114
Mobile Weight (grams)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
RAM Memory (GB)	0.0000	[0.0000, 0.0371]	0.0000	0.0095
Front Camera (MP)	0.0000	[-0.0257, 0.0166]	0.0000	0.0108
Max Back Camera MP	0.0000	[-0.0312, 0.0000]	0.0000	0.0080
Number of Cameras	0.0000	[-0.0165, 0.0000]	0.0000	0.0042
Processor Level (encoded)	0.0000	[-0.0285, 0.0000]	0.0000	0.0073
Battery Capacity (mAh) [SELECTED]	0.0029	[0.0000, 0.0282]	0.8976	0.0072
Screen Size (inches)	0.0000	[0.0000, 0.0167]	0.0000	0.0043

Sample Size: 116 R^2 Score: 0.0867   Optimal Alpha: 0.011148   Features Selected: 2/9

[SELECTED] = Feature selected by Lasso   Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 52: Price-Change Regression Summary 2024 Q1 → 2024 Q2

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Mobile Weight (grams)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
RAM Memory (GB)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Front Camera (MP) [SELECTED]	0.0042	[0.0000, 0.0595]	0.9302	0.0152
Max Back Camera MP	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Number of Cameras	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Processor Level (encoded)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Battery Capacity (mAh)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Screen Size (inches)	0.0000	[0.0000, 0.0000]	0.0000	0.0000

Sample Size: 121 R^2 Score: 0.0340 Optimal Alpha: 0.027767 Features Selected: 1/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 53: Price-Change Regression Summary 2024 Q2 → 2024 Q3

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[0.0000, 0.0179]	0.0000	0.0046
Mobile Weight (grams)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
RAM Memory (GB)	0.0000	[-0.0159, 0.0000]	0.0000	0.0041
Front Camera (MP)	0.0000	[-0.0179, 0.0000]	0.0000	0.0046
Max Back Camera MP	0.0000	[-0.0279, 0.0000]	0.0000	0.0071
Number of Cameras	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Processor Level (encoded)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Battery Capacity (mAh)	0.0000	[-0.0374, 0.0000]	0.0000	0.0095
Screen Size (inches)	0.0000	[0.0000, 0.0000]	0.0000	0.0000

Sample Size: 127 R^2 Score: 0.0000 Optimal Alpha: 0.025265 Features Selected: 0/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 54: Price-Change Regression Summary 2024 Q3 → 2024 Q4

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	-0.0530***	[-0.1086, -0.0326]	0.0100	0.0194
Mobile Weight (grams) [SELECTED]	-0.0074	[-0.0410, 0.0102]	0.8550	0.0130
RAM Memory (GB) [SELECTED]	-0.0041	[-0.0629, 0.0396]	0.9601	0.0261
Front Camera (MP) [SELECTED]	-0.0184	[-0.0538, 0.0000]	0.6579	0.0137
Max Back Camera MP [SELECTED]	-0.0103	[-0.0405, 0.0000]	0.7456	0.0103
Number of Cameras [SELECTED]	0.0104	[0.0000, 0.0357]	0.7077	0.0091
Processor Level (encoded) [SELECTED]	-0.0245	[-0.0694, 0.0151]	0.7097	0.0216
Battery Capacity (mAh)	0.0000	[-0.0400, 0.0505]	0.0000	0.0231
Screen Size (inches) [SELECTED]	-0.0180	[-0.0522, 0.0000]	0.6552	0.0133

Sample Size: 128 R^2 Score: 0.1879 Optimal Alpha: 0.003126 Features Selected: 8/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 55: Price-Change Regression Summary 2024 Q4 → 2025 Q1

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0) [SELECTED]	-0.0244***	[-0.0466, 0.0000]	0.5000	0.0119
Mobile Weight (grams)	0.0000	[0.0000, 0.0000]	0.0000	0.0000
RAM Memory (GB) [SELECTED]	0.0034	[0.0000, 0.0402]	0.9149	0.0103
Front Camera (MP) [SELECTED]	0.0040	[0.0000, 0.0356]	0.8881	0.0091
Max Back Camera MP	0.0000	[0.0000, 0.0218]	0.0000	0.0055
Number of Cameras	0.0000	[0.0000, 0.0106]	0.0000	0.0027
Processor Level (encoded)	0.0000	[-0.0365, 0.0328]	0.0000	0.0177
Battery Capacity (mAh)	0.0000	[-0.0042, 0.0145]	0.0000	0.0048
Screen Size (inches) [SELECTED]	0.0010	[0.0000, 0.0222]	0.9542	0.0057

Sample Size: 138 R^2 Score: 0.1262 Optimal Alpha: 0.010523 Features Selected: 4/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2.5·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE

Table 56: Price-Change Regression Summary: 2025 Q1 → 2025 Q2

Feature	Coefficient	95% CI	P-Value	Std Error
iOS (1) vs Android (0)	0.0000	[-0.0047, 0.0188]	0.0000	0.0060
Mobile Weight (grams)	0.0000	[0.0000, 0.0147]	0.0000	0.0038
RAM Memory (GB) [SELECTED]	0.0085	[0.0000, 0.0503]	0.8315	0.0128
Front Camera (MP)	0.0000	[0.0000, 0.0211]	0.0000	0.0054
Max Back Camera MP	0.0000	[-0.0297, 0.0000]	0.0000	0.0076
Number of Cameras	0.0000	[0.0000, 0.0000]	0.0000	0.0000
Processor Level (encoded)	0.0000	[-0.0206, 0.0183]	0.0000	0.0099
Battery Capacity (mAh)	0.0000	[-0.0109, 0.0000]	0.0000	0.0028
Screen Size (inches) [SELECTED]	0.0007	[0.0000, 0.0214]	0.9668	0.0055

Sample Size: 131 R^2 Score: 0.0439 Optimal Alpha: 0.008865 Features Selected: 2/9

[SELECTED] = Feature selected by Lasso Significance: *** |coef| > 2·SE, ** |coef| > 1.96·SE, * |coef| > 1.645·SE